

MTH-165: Mathematics for Engineers

The Definitive University Exam Companion & Solved
Question Paper Book

Complete 16-Paper Assessment Suite (Sets A, B, C & D)

Featuring Step-by-Step Mathematical
Solutions & Examiner Rubrics

4 Full CA-1 Papers (Units 1 & 2)
4 Full CA-2 Papers (Units 4 & 5)

4 Full Mid-Term Papers (Units 1, 2,
4 Full End-Term Papers (Units 1 to

Prepared for Lovely Professional University (LPU) Engineering Students

Course Outcomes CO1–CO6 • Bloom's Taxonomy L1–L4 • Exact Marking Rubrics

LPU Engineering Open Series

Academic Session 2024–2025 / 2025–2026

Official Examination Architecture & Blueprint

This Examination Companion provides verified model question papers and complete step-by-step solutions designed strictly to mirror the latest Lovely Professional University (LPU) examination structure for **MTH165: Mathematics for Engineers**:

Table 1: **MTH165 Examination Structure and Marking Scheme**

Assessment	Syllabus Scope	Format	Marks	Instructions & Rules
CA-1	Unit 1 & Unit 2	6 Subjective Qs	30	Attempt any 5 out of 6 questions ($5 \times 6M = 30M$ or $5 \times 5M + 5M$ step grading). Time: 60 mins.
Mid-Term	Unit 1, Unit 2, Unit 3	30 MCQs	30	10 MCQs per unit. Negative marking -0.25 marks for wrong answers. Time: 90 mins.
CA-2	Unit 4 & Unit 5	6 Subjective Qs	30	Attempt any 5 out of 6 questions ($5 \times 6M = 30M$). Time: 60 mins.
End-Term	Units 1 to 6	Part A: 30 MCQs Part B: 5 Subj Qs	70	Part A: 30 MCQs from Units 4, 5, 6 (10 each). Part B: 5 Qs of 10M (3 from U4-6, 2 from U1-6), attempt any 4 ($4 \times 10 = 40M$). Time: 3 Hours.

Master Formula Sheet: Units 1 to 6

Unit 1: Matrices & Linear Algebra Unit 3: Infinite Series

- **Characteristic Eqn:** $\det(A - \lambda I) = 0$
- For 3×3 matrix: $\lambda^3 - \text{tr}(A)\lambda^2 + (M_{11} + M_{22} + M_{33})\lambda - \det(A) = 0$
- **Cayley-Hamilton:** Matrix A satisfies its own characteristic equation: $P(A) = O$.
- $\sum \lambda_i = \text{trace}(A)$, $\prod \lambda_i = \det(A)$.
- Eigenvalues of A^{-1} are $1/\lambda_i$; eigenvalues of A^k are λ_i^k .
- **System $AX = B$ Consistency:**
 - $\text{rank}(A) \neq \text{rank}[A|B] \implies$ Inconsistent (No solution).
 - $\text{rank}(A) = \text{rank}[A|B] = n \implies$ Unique solution.
 - $\text{rank}(A) = \text{rank}[A|B] < n \implies$ Infinitely many solutions ($n - r$ arbitrary constants).
- **Hermitian Matrix:** $A^\dagger = (\bar{A})^T = A$. (Eigenvalues are strictly real).
- **Skew-Hermitian:** $A^\dagger = -A$. (Eigenvalues are zero or purely imaginary).
- **Orthogonal:** $A^T A = I \implies \det(A) = \pm 1$.
- **p-Series Test:** $\sum \frac{1}{n^p}$ converges if $p > 1$, diverges if $p \leq 1$.
- **D'Alembert's Ratio Test:** $L = \lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n}$. $L < 1$ converges, $L > 1$ diverges, $L = 1$ inconclusive.
- **Cauchy's Root Test:** $L = \lim_{n \rightarrow \infty} (u_n)^{1/n}$.

Unit 2: Differential Equations

- General solution: $y = y_c + y_p$
- Roots of auxiliary equation $f(m) = 0$:
 - Real & distinct: $c_1 e^{m_1 x} + c_2 e^{m_2 x}$
 - Repeated: $(c_1 + c_2 x) e^{mx}$
 - Complex $\alpha \pm i\beta$: $e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)$
- **Particular Integral (P.I.):**
 - $\frac{1}{f(D)} e^{ax} = \frac{e^{ax}}{f(a)}$ if $f(a) \neq 0$.
 - $\frac{1}{D^2 + a^2} \sin ax = -\frac{x}{2a} \cos ax$.
 - $\frac{1}{D^2 + a^2} \cos ax = \frac{x}{2a} \sin ax$.
 - $\frac{1}{f(D)} e^{ax} V = e^{ax} \frac{1}{f(D+a)} V$.
 - $\frac{1}{f(D)} x^m = [f(D)]^{-1} x^m$ (expand in powers of D).
- **Raabe's Test:** $L = \lim_{n \rightarrow \infty} n \left(\frac{u_n}{u_{n+1}} - 1 \right)$. $L > 1$ conv., $L < 1$ div.
- **Leibniz's Test (Alternating):** $\sum (-1)^{n-1} u_n$ converges if $u_{n+1} \leq u_n$ and $\lim u_n = 0$.

Unit 4: Multivariable Calculus

- **Limits in $\mathbb{R}^2$:** If limits along $y = mx$ or $y = mx^2$ depend on m , the limit does not exist.
- **Schwarz's Theorem:** If f_{xy} and f_{yx} are continuous, then $f_{xy} = f_{yx}$.
- **Euler's Theorem:** If $f(x, y)$ is homogeneous of degree n : $x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = n f(x, y)$.
- **Jacobian:** $J = \frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix}$.
- **Chain Rule for Jacobians:** $J \cdot J' = 1$.
- **Maxima/Minima in $\mathbb{R}^2$:** Let $r = f_{xx}$, $s = f_{xy}$, $t = f_{yy}$ at stationary point (a, b) where $f_x = f_y = 0$:
 - $rt - s^2 > 0$ and $r < 0 \implies$ Local Maximum.
 - $rt - s^2 > 0$ and $r > 0 \implies$ Local Minimum.
 - $rt - s^2 < 0 \implies$ Saddle Point.
 - $rt - s^2 = 0 \implies$ Inconclusive test.
- **Lagrange Multipliers:** $\nabla f = \lambda \nabla g$.

Unit 5: Multiple Integrals

- **Cartesian to Polar:** $x = r \cos \theta, y = r \sin \theta, dx dy = r dr d\theta$.

- **Change of Order of Integration:** Sketch the domain, reverse slicing direction (vertical strips $\leftrightarrow$ horizontal strips).
- **Cylindrical Coordinates:** $x = r \cos \theta, y = r \sin \theta, z = z, dV = r dz dr d\theta$.
- **Spherical Coordinates:** $x = \rho \sin \phi \cos \theta, y = \rho \sin \phi \sin \theta, z = \rho \cos \phi, dV = \rho^2 \sin \phi d\rho d\phi d\theta$.

Unit 6: Vector Calculus

- $\nabla \phi = \frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k}$ (Normal vector).
- **Directional Derivative:** $D_{\hat{u}} \phi = \nabla \phi \cdot \hat{u}$.
- **Divergence:** $\nabla \cdot \vec{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$. (Solenoidal if $\nabla \cdot \vec{F} = 0$).
- **Curl:** $\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix}$. (Irrotational if $\nabla \times \vec{F} = \vec{0}$).
- **Green's Theorem:** $\oint_C (P dx + Q dy) = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$.
- **Stokes' Theorem:** $\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \hat{n} dS$.
- **Gauss Divergence Theorem:** $\iint_S \vec{F} \cdot \hat{n} dS = \iiint_V (\nabla \cdot \vec{F}) dV$.

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Chapter 1

Continuous Assessment 1 (CA-1) Suite

Continuous Assessment 1 evaluates students on **Unit 1 (Matrix Methods and Linear Systems)** and **Unit 2 (Differential Calculus and Applications)**. Each paper contains **6 questions** carrying **5 marks each**, and students attempt **5 questions** (Total = **30 Marks**, Time Allowed = **60 Minutes**).

1.1 Continuous Assessment 1 (CA-1) — Set A

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Course Code: MTH-165 **Course Title:** Mathematics for Engineers**Academic Task Type:** Continuous Assessment (CA-1) **Set:** A**Maximum Marks:** 30 Marks **Time Allowed:** 50 Minutes**Roll No.:****Reg. No.:****Section:****Course Outcomes Addressed:** CO1 (Matrices & Systems of Linear Equations), CO2 (Higher-Order ODEs & Calculus)**Note: Attempt all six questions. Each question carries 5 marks.**

- Q.1** Find the characteristic equation and compute the eigenvalues of the matrix:

CO1

L2 Under-stand

5 Marks

$$A = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 2 & 1 \\ -1 & 2 & 2 \end{bmatrix}$$

- Q.2** If the rank of the matrix A is 1, determine the value of the non-zero constant a :

CO1

L3 Apply

5 Marks

$$A = \begin{bmatrix} a & -a & 0 \\ 0 & 1 & 0 \\ -a & 0 & 1 \end{bmatrix}$$

Can $\text{rank}(A) = 1$ be achieved for any real value of a ? Justify rigorously.

- Q.3** Determine the values of parameters a and b such that the following non-homogeneous system of linear equations possesses a **unique solution**:

CO1

L3 Apply

5 Marks

$$\begin{aligned} x - 2y + z &= 1 \\ x + 4y + 3z &= -1 \\ x + 4y + az &= -b \end{aligned}$$

- Q.4** Test whether the following set of vectors in $\mathbb{R}^4$ is linearly dependent or linearly independent:

CO1

L3 Apply

5 Marks

$$v_1 = (0, -1, -1, 0), \quad v_2 = (0, 0, -1, 1), \quad v_3 = (1, -1, 1, 1)$$

- Q.5** Find the eigenvalues and corresponding eigenvectors of the upper triangular matrix:

CO1

L3 Apply

**5
Marks**

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- Q.6** Solve the third-order linear homogeneous ordinary differential equation by the operator method:

CO2

L3 Apply

**5
Marks**

$$3y''' - 2y'' - 3y' + 2y = 0$$

Solutions & Marking Scheme (CA-1 Set A)

Step-by-Step Mathematical Solution Solution to Question 1 (5 Marks)

Step 1: Formula for Characteristic Equation The characteristic equation of a 3×3 matrix A is given by:

$$\det(A - \lambda I) = \lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

where:

- $S_1 = \text{trace}(A) = 1 + 2 + 2 = 5$.
- $S_2 = \text{Sum of minors of principal diagonal elements:}$

$$M_{11} = \begin{vmatrix} 2 & 1 \\ 2 & 2 \end{vmatrix} = 4 - 2 = 2$$

$$M_{22} = \begin{vmatrix} 1 & 2 \\ -1 & 2 \end{vmatrix} = 2 - (-2) = 4$$

$$M_{33} = \begin{vmatrix} 1 & 2 \\ 0 & 2 \end{vmatrix} = 2 - 0 = 2$$

Hence, $S_2 = 2 + 4 + 2 = 8$.

- $S_3 = \det(A)$: Expanding along row 1:

$$\det(A) = 1(4 - 2) - 2(0 - (-1)) + 2(0 - (-2)) = 1(2) - 2(1) + 2(2) = 2 - 2 + 4 = 4$$

Step 2: Characteristic Equation

$$\lambda^3 - 5\lambda^2 + 8\lambda - 4 = 0$$

Step 3: Solving for Eigenvalues By inspection, test $\lambda = 1$:

$$1^3 - 5(1)^2 + 8(1) - 4 = 1 - 5 + 8 - 4 = 0 \implies (\lambda - 1) \text{ is a factor.}$$

Factorizing by synthetic division:

$$(\lambda - 1)(\lambda^2 - 4\lambda + 4) = 0 \implies (\lambda - 1)(\lambda - 2)^2 = 0$$

Thus, the eigenvalues are:

$$\lambda_1 = 1, \quad \lambda_2 = 2, \quad \lambda_3 = 2$$

Mark Distribution: • Characteristic equation formula & trace/minors: **2 Marks**

• Correct determinant & cubic equation $\lambda^3 - 5\lambda^2 + 8\lambda - 4 = 0$: **1.5 Marks**

• Correct eigenvalues $\lambda = 1, 2, 2$: **1.5 Marks**

Step-by-Step Mathematical Solution Solution to Question 2 (5 Marks)

Step 1: Understanding Rank of Matrix The rank of a matrix is the maximum number of linearly independent rows (or the order of the highest-order non-vanishing minor).

For $\text{rank}(A) = 1$, **all 2×2 minors must vanish simultaneously**. Let us evaluate minors of order 2 from:

$$A = \begin{bmatrix} a & -a & 0 \\ 0 & 1 & 0 \\ -a & 0 & 1 \end{bmatrix}$$

Look at the submatrix formed by rows 2 and 3, columns 2 and 3:

$$M = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = (1)(1) - (0)(0) = 1 \neq 0$$

Step 2: Analysis Since there exists a 2×2 minor that equals $1 \neq 0$ **independent of the value of a** , the rank of A must be at least 2 for all values of a . That is:

$$\text{rank}(A) \geq 2 \quad \forall a \in \mathbb{R}$$

Therefore, **no value of a exists** such that $\text{rank}(A) = 1$.

University Exam Twist

This is a standard trap question in LPU exams. Students often attempt to set $\det(A) = 0$ to get a , but $\det(A) = 0$ only guarantees $\text{rank}(A) \leq 2$, NOT $\text{rank}(A) = 1$. Always check 2×2 minors!

Mark Distribution: • Condition for rank = 1 (all 2×2 minors = 0): **2 Marks**

• Explicit computation of a non-zero 2×2 minor: **2 Marks**

• Conclusive statement that no such a exists: **1 Mark**

Step-by-Step Mathematical Solution Solution to Question 3 (5 Marks)

Step 1: Augmented Matrix Formulation The augmented matrix $[A|B]$ is:

$$[A|B] = \left[\begin{array}{ccc|c} 1 & -2 & 1 & 1 \\ 1 & 4 & 3 & -1 \\ 1 & 4 & a & -b \end{array} \right]$$

Step 2: Elementary Row Operations (Echelon Form) Apply $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$\left[\begin{array}{ccc|c} 1 & -2 & 1 & 1 \\ 0 & 6 & 2 & -2 \\ 0 & 6 & a-1 & -b-1 \end{array} \right]$$

Now apply $R_3 \rightarrow R_3 - R_2$:

$$\left[\begin{array}{ccc|c} 1 & -2 & 1 & 1 \\ 0 & 6 & 2 & -2 \\ 0 & 0 & a-3 & -b+1 \end{array} \right]$$

Step 3: Condition for Unique Solution (Rouché-Capelli Theorem) For the system to have a ****unique solution****, the rank of coefficient matrix A must equal the rank of the augmented matrix $[A|B]$ and must equal the number of unknowns ($n = 3$):

$$\text{rank}(A) = \text{rank}[A|B] = 3$$

This strictly requires the pivot element in the third row to be non-zero:

$$a - 3 \neq 0 \implies \mathbf{a \neq 3}$$

The value of b has no effect on $\text{rank}(A)$ when $a \neq 3$. Therefore:

$$\mathbf{a \neq 3} \quad \text{and} \quad \mathbf{b \in \mathbb{R}} \text{ (any real number)}$$

Mark Distribution: • Setting up augmented matrix: **1 Mark**

• Row reduction to row echelon form: **2 Marks**

• Correct condition $a \neq 3$ and $b \in \mathbb{R}$: **2 Marks**

Step-by-Step Mathematical Solution Solution to Question 4 (5 Marks)

Step 1: Linear Combination Setup Let $c_1, c_2, c_3 \in \mathbb{R}$ such that:

$$c_1 v_1 + c_2 v_2 + c_3 v_3 = \vec{0}$$

$$c_1(0, -1, -1, 0) + c_2(0, 0, -1, 1) + c_3(1, -1, 1, 1) = (0, 0, 0, 0)$$

Step 2: Component-wise Equations

$$\text{Component 1: } c_3 = 0 \quad (1.1)$$

$$\text{Component 2: } -c_1 - c_3 = 0 \quad (1.2)$$

$$\text{Component 3: } -c_1 - c_2 + c_3 = 0 \quad (1.3)$$

$$\text{Component 4: } c_2 + c_3 = 0 \quad (1.4)$$

Step 3: Solving the System From (1.1), $c_3 = 0$. Substituting $c_3 = 0$ into (1.2):

$$-c_1 - 0 = 0 \implies c_1 = 0$$

Substituting $c_3 = 0$ into (1.4):

$$c_2 + 0 = 0 \implies c_2 = 0$$

Check in (1.3): $-0 - 0 + 0 = 0$ (Consistent).

Since the only solution is the trivial solution:

$$c_1 = c_2 = c_3 = 0$$

The vectors $\{v_1, v_2, v_3\}$ are **Linearly Independent**.

Mark Distribution: • Vector equation setup $c_1 v_1 + c_2 v_2 + c_3 v_3 = \vec{0}$: **1.5 Marks**

• Matrix or algebraic equation system: **1.5 Marks**

• Proving $c_1 = c_2 = c_3 = 0$ and concluding linear independence: **2 Marks**

Step-by-Step Mathematical Solution Solution to Question 5 (5 Marks)

Step 1: Eigenvalues For an upper triangular matrix, the eigenvalues are simply the diagonal entries:

$$\det(A - \lambda I) = (1 - \lambda)^3 = 0 \implies \lambda_1 = \lambda_2 = \lambda_3 = 1$$

Thus, $\lambda = 1$ is a repeated eigenvalue with algebraic multiplicity $AM = 3$.

Step 2: Eigenvectors for $\lambda = 1$ Solve $(A - I)X = \vec{0}$:

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

This yields a single constraint:

$$0x_1 + 1x_2 + 0x_3 = 0 \implies \mathbf{x}_2 = \mathbf{0}$$

Here, x_1 and x_3 are completely free variables! Let $x_1 = s$ and $x_3 = t$ ($s, t \in \mathbb{R}$, not both zero):

$$X = \begin{bmatrix} s \\ 0 \\ t \end{bmatrix} = s \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Thus, there are two linearly independent eigenvectors:

$$\mathbf{X}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{X}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Geometric multiplicity $GM = 3 - \text{rank}(A - I) = 3 - 1 = 2$.

Mark Distribution: • Finding eigenvalue $\lambda = 1$ (repeated): **1 Mark**

• Setting up system $(A - I)X = 0$: **1.5 Marks**

• Identifying free variables and producing two linearly independent eigenvectors: **2.5 Marks**

Step-by-Step Mathematical Solution Solution to Question 6 (5 Marks)

Step 1: Differential Operator Form Let $D = \frac{d}{dx}$. The given ODE is:

$$(3D^3 - 2D^2 - 3D + 2)y = 0$$

Step 2: Auxiliary Equation Replace D by m :

$$3m^3 - 2m^2 - 3m + 2 = 0$$

Group terms:

$$m^2(3m - 2) - 1(3m - 2) = 0$$

$$(3m - 2)(m^2 - 1) = 0 \implies (3m - 2)(m - 1)(m + 1) = 0$$

The roots of the auxiliary equation are:

$$m_1 = \frac{2}{3}, \quad m_2 = 1, \quad m_3 = -1$$

Step 3: General Solution Since all three roots are real and distinct, the complementary function (which is the general solution since the RHS is 0) is:

$$y(x) = c_1 e^{\frac{2}{3}x} + c_2 e^x + c_3 e^{-x}$$

where c_1, c_2, c_3 are arbitrary constants.

- Mark Distribution:**
- Auxiliary equation formulation: **1.5 Marks**
 - Factorization and finding roots $m = 2/3, 1, -1$: **2 Marks**
 - Writing final general solution with arbitrary constants: **1.5 Marks**

1.2 Continuous Assessment 1 (CA-1) — Set B

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Course Code: MTH-165 **Course Title:** Mathematics for Engineers

Academic Task Type: Continuous Assessment (CA-1) **Set:** B

Maximum Marks: 30 Marks **Time Allowed:** 50 Minutes

Roll No.:

Reg. No.:

Section:

Course Outcomes Addressed: CO1 (Matrices & Systems of Linear Equations), CO2 (Higher-Order ODEs & Calculus)

Note: Attempt all six questions. Each question carries 5 marks.

Q.1 Find for what values of λ and μ the system of linear equations:

CO1

L3 Apply

**5
Marks**

$$\begin{aligned}x + y + z &= 6 \\x + 2y + 3z &= 10 \\x + 2y + \lambda z &= \mu\end{aligned}$$

has: (i) a unique solution, (ii) no solution, and (iii) infinitely many solutions.

Q.2 Solve the third-order linear homogeneous differential equation by the operator method:

CO2

L3 Apply

**5
Marks**

$$y''' + y = 0 \quad \Longleftrightarrow \quad (D^3 + 1)y = 0$$

Q.3 Find the rank of the 4×4 matrix A , given that parameters $w, r, s, t \neq 0$:

CO1

L2 Understand

**5
Marks**

$$A = \begin{bmatrix} 1 & w & r & 0 \\ 0 & s & t & 1 \\ -1 & -w & -r & 0 \\ 0 & s & t & 1 \end{bmatrix}$$

Q.4 Verify the **Cayley-Hamilton Theorem** for the matrix:

CO1

L3 Apply

**5
Marks**

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

and hence compute A^{-1} and A^3 .

Q.5

Determine whether the matrix:

CO1

L3 Apply

**5
Marks**

$$A = \begin{bmatrix} 2 & 3 + 4i \\ 3 - 4i & 5 \end{bmatrix}$$

is a **Hermitian matrix**. Find its eigenvalues and verify that they are strictly real.

Q.6

Find the complete general solution of the second-order differential equation:

CO2

L3 Apply

**5
Marks**

$$(D^2 - 4D + 4)y = e^{2x}$$

Solutions & Marking Scheme (CA-1 Set B)

Step-by-Step Mathematical Solution Solution to Question 1 (5 Marks)

Step 1: Augmented Matrix Formulation

$$[A|B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 10 \\ 1 & 2 & \lambda & \mu \end{array} \right]$$

Step 2: Row Echelon Reduction Apply $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 1 & \lambda - 1 & \mu - 6 \end{array} \right]$$

Apply $R_3 \rightarrow R_3 - R_2$:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & \lambda - 3 & \mu - 10 \end{array} \right]$$

Step 3: Classification of Solutions

- (i) **Unique Solution:** Requires $\text{rank}(A) = \text{rank}[A|B] = 3$ (number of unknowns). Hence, the pivot $\lambda - 3 \neq 0 \implies \lambda \neq 3$, and $\mu \in \mathbb{R}$ (any value).
- (ii) **No Solution (Inconsistent):** Requires $\text{rank}(A) = 2$ and $\text{rank}[A|B] = 3$. This occurs when $\lambda - 3 = 0$ and $\mu - 10 \neq 0$:

$$\lambda = 3 \quad \text{and} \quad \mu \neq 10$$

- (iii) **Infinitely Many Solutions:** Requires $\text{rank}(A) = \text{rank}[A|B] = 2 < 3$. This occurs when the entire third row becomes zero:

$$\lambda = 3 \quad \text{and} \quad \mu = 10$$

Mark Distribution: • Augmented matrix and row operations: **2 Marks**

- Unique solution condition ($\lambda \neq 3, \mu \in \mathbb{R}$): **1 Mark**
- No solution condition ($\lambda = 3, \mu \neq 10$): **1 Mark**
- Infinite solution condition ($\lambda = 3, \mu = 10$): **1 Mark**

Step-by-Step Mathematical Solution Solution to Question 2 (5 Marks)

Step 1: Auxiliary Equation Given $(D^3 + 1)y = 0$, replace D by m :

$$m^3 + 1 = 0$$

Use algebraic identity $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$:

$$(m + 1)(m^2 - m + 1) = 0$$

Step 2: Finding Roots

- From $m + 1 = 0 \Rightarrow \mathbf{m_1 = -1}$.
- From $m^2 - m + 1 = 0$, using quadratic formula:

$$m = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(1)(1)}}{2(1)} = \frac{1 \pm \sqrt{-3}}{2} = \frac{1}{2} \pm i \frac{\sqrt{3}}{2}$$

Step 3: General Solution For the real root $m_1 = -1$, the term is $c_1 e^{-x}$. For complex conjugate roots $\alpha \pm i\beta$ with $\alpha = 1/2, \beta = \sqrt{3}/2$:

$$e^{\frac{1}{2}x} \left(c_2 \cos \frac{\sqrt{3}}{2}x + c_3 \sin \frac{\sqrt{3}}{2}x \right)$$

Therefore, the general solution is:

$$\mathbf{y(x) = c_1 e^{-x} + e^{\frac{x}{2}} \left(c_2 \cos \frac{\sqrt{3}}{2}x + c_3 \sin \frac{\sqrt{3}}{2}x \right)}$$

Mark Distribution: • Auxiliary equation $m^3 + 1 = 0$ and factorization: **1.5 Marks**

• Correct roots $m = -1, \frac{1}{2} \pm i \frac{\sqrt{3}}{2}$: **1.5 Marks**

• Final general solution with all 3 arbitrary constants: **2 Marks**

Step-by-Step Mathematical Solution Solution to Question 3 (5 Marks)**Step 1: Matrix Inspection & Row Operations**

$$A = \begin{bmatrix} 1 & w & r & 0 \\ 0 & s & t & 1 \\ -1 & -w & -r & 0 \\ 0 & s & t & 1 \end{bmatrix}$$

Notice that:

- Row 3 is the exact negative of Row 1: $R_3 = -R_1$.
- Row 4 is identical to Row 2: $R_4 = R_2$.

Step 2: Row Reduction to Echelon Form Apply $R_3 \rightarrow R_3 + R_1$ and $R_4 \rightarrow R_4 - R_2$:

$$\begin{bmatrix} 1 & w & r & 0 \\ 0 & s & t & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Step 3: Determining Rank Since $s \neq 0$, the first two rows are non-zero and linearly independent. The remaining two rows are entirely zero. The number of non-zero rows in the row echelon form is 2. Therefore:

$$\text{rank}(\mathbf{A}) = 2$$

Mark Distribution: • Identifying linear dependency between rows: **2 Marks**

• Performing row operations to echelon form: **2 Marks**

• Correct rank = 2: **1 Mark**

Step-by-Step Mathematical Solution Solution to Question 4 (5 Marks)

Step 1: Characteristic Equation of A For $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$:

$$\text{trace}(A) = 1 + 4 = 5, \quad \det(A) = (1)(4) - (2)(3) = 4 - 6 = -2$$

The characteristic equation is:

$$\lambda^2 - 5\lambda - 2 = 0$$

Step 2: Verifying Cayley-Hamilton Theorem The theorem asserts that A satisfies its own characteristic equation:

$$A^2 - 5A - 2I = O$$

Compute A^2 :

$$A^2 = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 1+6 & 2+8 \\ 3+12 & 6+16 \end{bmatrix} = \begin{bmatrix} 7 & 10 \\ 15 & 22 \end{bmatrix}$$

Now compute $A^2 - 5A - 2I$:

$$\begin{bmatrix} 7 & 10 \\ 15 & 22 \end{bmatrix} - 5 \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} - 2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 7-5-2 & 10-10-0 \\ 15-15-0 & 22-20-2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O$$

Hence, Cayley-Hamilton Theorem is verified!

Step 3: Calculating A^{-1} using Cayley-Hamilton Multiply the verified relation by A^{-1} :

$$A - 5I - 2A^{-1} = O \implies 2A^{-1} = A - 5I \implies A^{-1} = \frac{1}{2}(A - 5I)$$

$$A^{-1} = \frac{1}{2} \left(\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} - \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} \right) = \frac{1}{2} \begin{bmatrix} -4 & 2 \\ 3 & -1 \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 3/2 & -1/2 \end{bmatrix}$$

Mark Distribution: • Characteristic equation $\lambda^2 - 5\lambda - 2 = 0$: **1.5 Marks**

• Verification $A^2 - 5A - 2I = O$: **2 Marks**

• Computation of A^{-1} : **1.5 Marks**

Step-by-Step Mathematical Solution Solution to Question 5 (5 Marks)

Step 1: Definition and Verification of Hermitian Matrix A complex square matrix A is Hermitian if $A^\dagger = (\bar{A})^T = A$. Given:

$$A = \begin{bmatrix} 2 & 3+4i \\ 3-4i & 5 \end{bmatrix}$$

Take the complex conjugate $\bar{A}$:

$$\bar{A} = \begin{bmatrix} 2 & 3-4i \\ 3+4i & 5 \end{bmatrix}$$

Transpose $(\bar{A})^T$:

$$(\bar{A})^T = \begin{bmatrix} 2 & 3+4i \\ 3-4i & 5 \end{bmatrix} = A$$

Thus, A is indeed a Hermitian matrix.

Step 2: Eigenvalues Computation Characteristic equation: $\det(A - \lambda I) = 0$:

$$\begin{vmatrix} 2-\lambda & 3+4i \\ 3-4i & 5-\lambda \end{vmatrix} = 0$$

$$(2-\lambda)(5-\lambda) - (3+4i)(3-4i) = 0$$

$$(\lambda^2 - 7\lambda + 10) - (3^2 - (4i)^2) = 0$$

$$(\lambda^2 - 7\lambda + 10) - (9 + 16) = 0 \implies \lambda^2 - 7\lambda - 15 = 0$$

Solve for λ :

$$\lambda = \frac{7 \pm \sqrt{(-7)^2 - 4(1)(-15)}}{2} = \frac{7 \pm \sqrt{49 + 60}}{2} = \frac{7 \pm \sqrt{109}}{2}$$

Since $109 > 0$, both eigenvalues $\lambda_1 = \frac{7+\sqrt{109}}{2} \approx 8.72$ and $\lambda_2 = \frac{7-\sqrt{109}}{2} \approx -1.72$ are **strictly real**. This confirms the fundamental theorem that every Hermitian matrix has real eigenvalues.

Mark Distribution: • Proving $(\bar{A})^T = A$ for Hermitian: **2 Marks**

• Characteristic equation $\lambda^2 - 7\lambda - 15 = 0$: **1.5 Marks**

• Finding real eigenvalues: **1.5 Marks**

Step-by-Step Mathematical Solution Solution to Question 6 (5 Marks)**Step 1: Complementary Function (y_c)** Auxiliary equation:

$$m^2 - 4m + 4 = 0 \implies (m - 2)^2 = 0 \implies m = 2, 2 \text{ (repeated real root)}$$

$$y_c = (c_1 + c_2 x)e^{2x}$$

Step 2: Particular Integral (y_p)

$$y_p = \frac{1}{(D - 2)^2} e^{2x}$$

Notice that substituting $D = 2$ gives $(2 - 2)^2 = 0$ (Case of failure of order 2). Using the standard operator rule $\frac{1}{(D-a)^n} e^{ax} = \frac{x^n}{n!} e^{ax}$. Here $a = 2, n = 2$:

$$y_p = \frac{x^2}{2!} e^{2x} = \frac{1}{2} x^2 e^{2x}$$

Step 3: General Solution

$$y = y_c + y_p = (c_1 + c_2 x)e^{2x} + \frac{1}{2} x^2 e^{2x}$$

Mark Distribution: • Complementary function $y_c = (c_1 + c_2 x)e^{2x}$: **2 Marks**• Case of failure recognition and Particular Integral y_p : **2 Marks**• Final general solution $y = y_c + y_p$: **1 Mark**

1.3 Continuous Assessment 1 (CA-1) — Set C

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Course Code: MTH-165 **Course Title:** Mathematics for Engineers

Academic Task Type: Continuous Assessment (CA-1) **Set:** C

Maximum Marks: 30 Marks **Time Allowed:** 50 Minutes

Roll No.:

Reg. No.:

Section:

Course Outcomes Addressed: CO1 (Matrices & Systems of Linear Equations), CO2 (Higher-Order ODEs & Calculus)

Note: Attempt all six questions. Each question carries 5 marks.

Q.1 Find the rank and nullity of the 3×3 matrix:

CO1

L3 Apply

**5
Marks**

$$A = \begin{bmatrix} 2 & -4 & 6 \\ -1 & 2 & -3 \\ 3 & -6 & 9 \end{bmatrix}$$

Verify the **Rank-Nullity Theorem**.

Q.2 Find the inverse of the matrix B using the **Gauss-Jordan elimination method**:

CO1

L3 Apply

**5
Marks**

$$B = \begin{bmatrix} 2 & 4 & 2 \\ 6 & 4 & 6 \\ 2 & 4 & 4 \end{bmatrix}$$

Q.3 Solve the second-order differential equation with a resonance term:

CO2

L3 Apply

**5
Marks**

$$(D^2 + 4)y = \sin 2x$$

Q.4 Solve the **Cauchy-Euler homogeneous linear differential equation**:

CO2

L3 Apply

**5
Marks**

$$x^2 \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} + 2y = x^3$$

Q.5 Using the **Method of Variation of Parameters**, find the complete general solution of:

CO2

L4 Analyze

**5
Marks**

$$\frac{d^2 y}{dx^2} + y = \sec x$$

Q.6 Prove that all eigenvalues of a real **skew-symmetric matrix** are either zero or purely imaginary complex numbers. **CO1**

L4 Analyze **5 Marks**

Solutions & Marking Scheme (CA-1 Set C)

Step-by-Step Mathematical Solution Solution to Question 1 (5 Marks)

Step 1: Row Reduction to Echelon Form Given:

$$A = \begin{bmatrix} 2 & -4 & 6 \\ -1 & 2 & -3 \\ 3 & -6 & 9 \end{bmatrix}$$

Divide Row 1 by 2:

$$\begin{bmatrix} 1 & -2 & 3 \\ -1 & 2 & -3 \\ 3 & -6 & 9 \end{bmatrix}$$

Apply $R_2 \rightarrow R_2 + R_1$ and $R_3 \rightarrow R_3 - 3R_1$:

$$\begin{bmatrix} 1 & -2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Step 2: Rank and Nullity The echelon form contains exactly 1 non-zero row. Therefore:

$$\text{rank}(\mathbf{A}) = 1$$

The number of columns (dimension of domain) is $n = 3$. By the Rank-Nullity Theorem:

$$\text{rank}(A) + \text{nullity}(A) = n \implies 1 + \text{nullity}(A) = 3 \implies \text{nullity}(\mathbf{A}) = 2$$

Step 3: Verification $\text{rank}(A) + \text{nullity}(A) = 1 + 2 = 3 = \text{dimension of space}$. Hence verified.

Mark Distribution: • Row reduction to echelon form: **2 Marks**

• Correct rank = 1: **1 Mark**

• Correct nullity = 2 and Rank-Nullity Theorem verification: **2 Marks**

Step-by-Step Mathematical Solution Solution to Question 2 (5 Marks)**Step 1: Gauss-Jordan Augmented Matrix $[B|I]$**

$$[B|I] = \left[\begin{array}{ccc|ccc} 2 & 4 & 2 & 1 & 0 & 0 \\ 6 & 4 & 6 & 0 & 1 & 0 \\ 2 & 4 & 4 & 0 & 0 & 1 \end{array} \right]$$

Step 2: Row Operations $R_1 \rightarrow \frac{1}{2}R_1$:

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1/2 & 0 & 0 \\ 6 & 4 & 6 & 0 & 1 & 0 \\ 2 & 4 & 4 & 0 & 0 & 1 \end{array} \right]$$

 $R_2 \rightarrow R_2 - 6R_1$ and $R_3 \rightarrow R_3 - 2R_1$:

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1/2 & 0 & 0 \\ 0 & -8 & 0 & -3 & 1 & 0 \\ 0 & 0 & 2 & -1 & 0 & 1 \end{array} \right]$$

 $R_2 \rightarrow -\frac{1}{8}R_2$ and $R_3 \rightarrow \frac{1}{2}R_3$:

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1/2 & 0 & 0 \\ 0 & 1 & 0 & 3/8 & -1/8 & 0 \\ 0 & 0 & 1 & -1/2 & 0 & 1/2 \end{array} \right]$$

 $R_1 \rightarrow R_1 - R_3$:

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & -1/2 \\ 0 & 1 & 0 & 3/8 & -1/8 & 0 \\ 0 & 0 & 1 & -1/2 & 0 & 1/2 \end{array} \right]$$

 $R_1 \rightarrow R_1 - 2R_2$:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1/4 & 1/4 & -1/2 \\ 0 & 1 & 0 & 3/8 & -1/8 & 0 \\ 0 & 0 & 1 & -1/2 & 0 & 1/2 \end{array} \right]$$

Step 3: Result

$$\mathbf{B}^{-1} = \begin{bmatrix} 1/4 & 1/4 & -1/2 \\ 3/8 & -1/8 & 0 \\ -1/2 & 0 & 1/2 \end{bmatrix} = \frac{1}{8} \begin{bmatrix} 2 & 2 & -4 \\ 3 & -1 & 0 \\ -4 & 0 & 4 \end{bmatrix}$$

Mark Distribution: • Augmented matrix setup: **1 Mark**• Forward row reduction: **2 Marks**• Backward substitution to identity matrix: **1 Mark**• Correct B^{-1} : **1 Mark**

Step-by-Step Mathematical Solution Solution to Question 3 (5 Marks)

Step 1: Complementary Function $m^2 + 4 = 0 \implies m = \pm 2i$.

$$y_c = c_1 \cos 2x + c_2 \sin 2x$$

Step 2: Particular Integral (y_p) with Resonance

$$y_p = \frac{1}{D^2 + 4} \sin 2x$$

Substituting $D^2 = -2^2 = -4$ gives a denominator of zero (Resonance/Failure case). Using the formula $\frac{1}{D^2 + a^2} \sin ax = -\frac{x}{2a} \cos ax$: With $a = 2$:

$$y_p = -\frac{x}{2(2)} \cos 2x = -\frac{x}{4} \cos 2x$$

Step 3: General Solution

$$y(x) = c_1 \cos 2x + c_2 \sin 2x - \frac{x}{4} \cos 2x$$

Mark Distribution: • Complementary function y_c : **1.5 Marks**

- Detection of case of failure: **1.5 Marks**
- Correct derivation of $y_p = -(x/4) \cos 2x$: **1.5 Marks**
- Final general solution: **0.5 Mark**

Step-by-Step Mathematical Solution Solution to Question 4 (5 Marks)

Step 1: Transformation to Constant Coefficient ODE Let $x = e^z \implies z = \ln x$, and let $\theta = \frac{d}{dz}$. Then:

$$x \frac{dy}{dx} = \theta y, \quad x^2 \frac{d^2 y}{dx^2} = \theta(\theta - 1)y$$

Substituting into the equation:

$$[\theta(\theta - 1) - 2\theta + 2]y = (e^z)^3$$

$$(\theta^2 - 3\theta + 2)y = e^{3z}$$

Step 2: Solving the Transformed Equation Auxiliary equation: $m^2 - 3m + 2 = 0 \implies (m - 1)(m - 2) = 0 \implies m = 1, 2$.

$$y_c = c_1 e^z + c_2 e^{2z}$$

Particular integral:

$$y_p = \frac{1}{\theta^2 - 3\theta + 2} e^{3z} = \frac{e^{3z}}{3^2 - 3(3) + 2} = \frac{e^{3z}}{9 - 9 + 2} = \frac{1}{2} e^{3z}$$

So $y(z) = c_1 e^z + c_2 e^{2z} + \frac{1}{2} e^{3z}$. **Step 3: Transforming back to x** Since $e^z = x$:

$$y(x) = c_1 x + c_2 x^2 + \frac{1}{2} x^3$$

Mark Distribution: • Substitution $x = e^z$ and differential operator transformation: **1.5 Marks**

• Solution of auxiliary equation: **1.5 Marks**

• Particular integral calculation: **1 Mark**

• Final solution in terms of x : **1 Mark**

Step-by-Step Mathematical Solution Solution to Question 5 (5 Marks)

Step 1: Complementary Function & Wronskian For $y'' + y = 0$, $m^2 + 1 = 0 \implies m = \pm i$.

$$y_1 = \cos x, \quad y_2 = \sin x \implies y_c = c_1 \cos x + c_2 \sin x$$

The Wronskian $W(y_1, y_2)$ is:

$$W = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = \cos^2 x + \sin^2 x = 1 \neq 0$$

Step 2: Variation of Parameters Formulas Let $y_p = u(x)y_1 + v(x)y_2$, where $R(x) = \sec x$:

$$u(x) = - \int \frac{y_2 R(x)}{W} dx = - \int \frac{\sin x \sec x}{1} dx = - \int \tan x dx = \ln |\cos x|$$

$$v(x) = \int \frac{y_1 R(x)}{W} dx = \int \frac{\cos x \sec x}{1} dx = \int 1 dx = x$$

Step 3: Particular Integral and General Solution

$$y_p = (\ln |\cos x|) \cos x + x \sin x$$

$$y(\mathbf{x}) = c_1 \cos \mathbf{x} + c_2 \sin \mathbf{x} + \cos \mathbf{x} \ln |\cos \mathbf{x}| + \mathbf{x} \sin \mathbf{x}$$

Mark Distribution: • Complementary function and Wronskian $W = 1$: **1.5 Marks**

• Evaluation of integrals $u(x)$ and $v(x)$: **2 Marks**

• Final general solution $y = y_c + y_p$: **1.5 Marks**

Step-by-Step Mathematical Solution Solution to Question 6 (5 Marks)

Step 1: Mathematical Setup Let $A \in \mathbb{R}^{n \times n}$ be a real skew-symmetric matrix, i.e., $A^T = -A$ and all entries of A are real ($\bar{A} = A$). Let $\lambda \in \mathbb{C}$ be an eigenvalue of A with associated non-zero eigenvector $X \in \mathbb{C}^n$:

$$AX = \lambda X$$

Step 2: Conjugate Transpose Operations Multiply on the left by the conjugate transpose $X^\dagger = (\bar{X})^T$:

$$X^\dagger AX = \lambda X^\dagger X = \lambda \|X\|^2$$

Now take the conjugate transpose of both sides of (1.3):

$$(X^\dagger AX)^\dagger = (\lambda X^\dagger X)^\dagger$$

Using $(ABC)^\dagger = C^\dagger B^\dagger A^\dagger$:

$$X^\dagger A^\dagger X = \bar{\lambda} X^\dagger X$$

Since A is real, $A^\dagger = (\bar{A})^T = A^T$. Because A is skew-symmetric, $A^T = -A$:

$$X^\dagger(-A)X = \bar{\lambda} X^\dagger X \implies -X^\dagger AX = \bar{\lambda} X^\dagger X$$

Step 3: Equating and Conclusion Substitute $X^\dagger AX = \lambda X^\dagger X$:

$$-\lambda X^\dagger X = \bar{\lambda} X^\dagger X \implies (\lambda + \bar{\lambda}) X^\dagger X = 0$$

Since $X \neq \vec{0}$, $X^\dagger X = \|X\|^2 > 0$. Therefore:

$$\lambda + \bar{\lambda} = 0 \implies 2\text{Re}(\lambda) = 0 \implies \text{Re}(\lambda) = 0$$

Thus, the real part of λ is strictly zero. Hence, λ is either **zero** or **purely imaginary**.

Mark Distribution: • Definition of skew-symmetric and eigenvalue equation $AX = \lambda X$: **1.5 Marks**

• Applying conjugate transpose and property $A^\dagger = -A$: **2 Marks**

• Deducing $\lambda + \bar{\lambda} = 0 \implies \text{Re}(\lambda) = 0$: **1.5 Marks**

Chapter 2

Mid-Term Examination Suite (30 MCQs — 30 Marks)

The Mid-Term Examination is an objective assessment evaluating students across **Unit 1 (Matrices)**, **Unit 2 (Differential Calculus)**, and **Unit 3 (Integral Calculus)**. Each paper consists of **30 Multiple-Choice Questions (MCQs)** carrying **1 mark each** (10 from Unit 1, 10 from Unit 2, 10 from Unit 3) with a negative marking deduction of **0.25 marks** per incorrect answer (Total = **30 Marks**, Time Allowed = **90 Minutes**).

2.1 Mid-Term Examination — Set A

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Course Code: MTH-165 **Course Title:** Mathematics for Engineers

Academic Task Type: Mid-Term Examination (Objective MCQ)

Maximum Marks: 30 Marks **Time Allowed:** 90 Minutes

Course Outcomes Addressed: CO1 (Unit 1: Matrices), CO2 (Unit 2: ODEs & Calculus), CO3 (Unit 3: Infinite Series)

Roll No.:

Reg. No.:

Section:

Instructions: All 30 questions are compulsory. Each question carries 1 mark. A penalty of 0.25 marks is deducted for each incorrect response.

Q.1 The differential coefficient of $\log(\tan x)$ with respect to x is:

[1 Mark]

- (A) $2 \sec 2x$ (C) $2 \sec^2 x$
(B) $2 \csc 2x$ (D) $2 \csc^2 x$

Q.2 The value of the indefinite integral $\int \frac{dx}{x^2+2x+2}$ is equal to:

[1 Mark]

- (A) $\tan^{-1}(x+1) + C$ (C) $(x+1) \tan^{-1} x + C$
(B) $\tan^{-1} x + C$ (D) $\frac{1}{2} \tan^{-1}(x+1) + C$

Q.3 If $y = x^x$ for $x > 0$, then the derivative $\frac{dy}{dx}$ is:

[1 Mark]

- (A) $x^x \ln x$ (C) $2 + \ln x$
(B) $x^x(1 + \ln x)$ (D) x^{x-1}

Q.4 The derivative of $f(x) = \ln(x + \sqrt{x^2 + 1})$ with respect to x is:

[1 Mark]

- (A) $\sqrt{x^2 + 1}$ (C) $\frac{1}{\sqrt{x^2 + 1}}$
(B) $\frac{x}{\sqrt{x^2 + 1}}$ (D) $1 + \frac{x}{\sqrt{x^2 + 1}}$

Q.5 The value of the definite integral $\int_0^\pi \sin^2 x \, dx$ is:

[1 Mark]

- (A) 0 (C) π
(B) $\frac{\pi}{2}$ (D) $\frac{\pi}{4}$

Q.6 For a non-singular square matrix A of order 3 with $\det(A) = 3$, the value of $\det(\text{adj}(A))$ is: [1 Mark]

- (A) 3 (C) 27
(B) 9 (D) 81

Q.7 If A is an invertible square matrix, then $\det(A^{-1})$ is strictly equal to: [1 Mark]

- (A) $\frac{1}{\det(A)}$ (C) $-\frac{1}{\det(A)}$
(B) $\det(A)$ (D) $-\det(A)$

Q.8 If 2, 3, -4 are the eigenvalues of a matrix A , then the eigenvalues of the matrix A^2 are: [1 Mark]

- (A) 2, 3, -4 (C) 4, 9, 16
(B) -2 , -3 , 4 (D) 4, 3, 16

Q.9 The rank of the upper triangular matrix $A = \begin{bmatrix} 0 & 2 & -1 \\ 1 & 5 & 2 \\ 0 & 0 & 7 \end{bmatrix}$ is: [1 Mark]

- (A) 1 (C) 3
(B) 2 (D) 0

Q.10 For any non-singular matrix A of order n , the matrix $\text{adj}(\text{adj}(A))$ is equal to: [1 Mark]

- (A) $|A|^{n-1}A$ (C) $|A|^nA$
(B) $|A|^{n-2}A$ (D) $|A|^{n-1}I$

Q.11 The value of $\int \frac{e^{\tan^{-1}x}}{1+x^2} dx$ is equal to: [1 Mark]

- (A) $e^{\tan^{-1}x} + C$ (C) $\frac{(\tan^{-1}x)^2}{2} + C$
(B) $\frac{(e^{\tan^{-1}x})^2}{2} + C$ (D) $\tan^{-1}(e^x) + C$

Q.12 The definite integral $\int_{-1}^1 (x^3 + x) dx$ evaluates to: [1 Mark]

- (A) 0 (C) π
(B) $\frac{\pi}{2}$ (D) 1

Q.13 If $x^y = e^{x-y}$, then $\frac{dy}{dx}$ is equal to: [1 Mark]

- (A) $\frac{y}{(1+\ln x)^2}$ (C) $\frac{x}{(1+\ln x)^2}$
(B) $\frac{\ln x}{(1+\ln x)^2}$ (D) $\frac{1}{(1+\ln x)^2}$

Q.14 The eigenvalues of a Hermitian matrix are always: [1 Mark]

- (A) All Real (C) Zero
(B) All Purely Imaginary (D) Unitary

Q.15 If A is an $m \times n$ matrix where $m < n$, then the maximum possible value of $\text{rank}(A)$ is: [1 Mark]

- (A) n (C) $m \cdot n$
(B) m (D) $n - m$

Q.16 The value of $\int \frac{e^x(1+x)}{\cos^2(xe^x)} dx$ is equal to: [1 Mark]

- (A) $-\cot(xe^x) + C$ (C) $\sec(xe^x) + C$
(B) $\tan(xe^x) + C$ (D) $\tan(e^x) + C$

Q.17 The system of linear equations $AX = B$ is **inconsistent** (has no solution) if: [1 Mark]

- (A) $\text{rank}(A) = \text{rank}[A|B] = n$ (C) $\text{rank}(A) = \text{rank}[A|B] < n$
(B) $\text{rank}(A) \neq \text{rank}[A|B]$ (D) $\det(A) \neq 0$

Q.18 The roots of the auxiliary equation for $(D^2 + 9)y = 0$ are: [1 Mark]

- (A) ± 3 (C) $3, 3$
(B) $\pm 3i$ (D) $-3, -3$

Q.19 The particular integral (P.I.) of $(D - 2)^2 y = e^{2x}$ is: [1 Mark]

- (A) $\frac{x^2}{2}e^{2x}$ (C) $\frac{x}{2}e^{2x}$
(B) xe^{2x} (D) $\frac{x^3}{6}e^{2x}$

Q.20 The general solution of the Cauchy-Euler equation $x^2 y'' - 2xy' + 2y = 0$ is: [1 Mark]

- (A) $c_1 x + c_2 x^2$ (C) $(c_1 + c_2 \ln x)x$
(B) $c_1 e^x + c_2 e^{2x}$ (D) $c_1 x + c_2 x^{-2}$

Q.21 The infinite p -series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is **convergent** if and only if: [1 Mark]

- (A) $p \leq 1$ (C) $p > 1$
(B) $p < 1$ (D) $p \geq 0$

Q.22 Using D'Alembert's Ratio Test, the series $\sum_{n=1}^{\infty} u_n$ converges if $\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = L$ satisfies: [1 Mark]

- (A) $L > 1$ (C) $L = 1$
(B) $L < 1$ (D) $L = \infty$

Q.23 The alternating series $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$ is: [1 Mark]

- (A) Divergent (C) Conditionally Convergent
(B) Absolutely Convergent (D) Oscillatory

Q.24 The radius of convergence R of the power series $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ is: [1 Mark]

- (A) 0 (C) e
(B) 1 (D) ∞

Q.25 The sum of the eigenvalues of the matrix $A = \begin{bmatrix} 4 & 2 & 1 \\ 0 & 3 & 5 \\ 0 & 0 & -2 \end{bmatrix}$ is: [1 Mark]

- (A) 5 (C) -24
(B) 7 (D) 10

Q.26 The product of the eigenvalues of the matrix $A = \begin{bmatrix} 2 & 1 \\ 4 & 5 \end{bmatrix}$ is: [1 Mark]

- (A) 6 (C) 10
(B) 7 (D) 8

Q.27 The series $\sum_{n=1}^{\infty} \frac{1}{n^2+1}$ is: [1 Mark]

- (A) Convergent by Comparison with $\sum \frac{1}{n^2}$ (C) Oscillatory
(B) Divergent by Comparison with $\sum \frac{1}{n}$ (D) Conditionally Convergent

Q.28 If A is an orthogonal matrix, then $\det(A)$ is always: [1 Mark]

- (A) 0 (C) ∞
(B) ± 1 (D) Any real number

Q.29 The Wronskian $W(y_1, y_2)$ of the functions $y_1 = \cos 2x$ and $y_2 = \sin 2x$ is: [1 Mark]

- (A) 1 (C) -2
(B) 2 (D) 0

Q.30 The particular integral of $(D^2 + 4)y = \cos 2x$ is: [1 Mark]

- (A) $-\frac{x}{4} \sin 2x$ (C) $\frac{x}{2} \cos 2x$
(B) $\frac{x}{4} \sin 2x$ (D) $-\frac{x}{2} \sin 2x$

Answer Key (Mid-Term Set A)

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	6	B	11	A	16	B	21	C	26	A
2	A	7	A	12	A	17	B	22	B	27	A
3	B	8	C	13	B	18	B	23	C	28	B
4	C	9	C	14	A	19	A	24	D	29	B
5	B	10	B	15	B	20	A	25	A	30	B

Selected Detailed Mathematical Explanations

Step-by-Step Mathematical Solution Explanations for Critical Questions

Q.1 (Ans: B) $\frac{d}{dx}[\ln(\tan x)] = \frac{1}{\tan x} \sec^2 x = \frac{\cos x}{\sin x} \frac{1}{\cos^2 x} = \frac{1}{\sin x \cos x} = \frac{2}{2 \sin x \cos x} = \frac{2}{\sin 2x} = 2 \csc 2x$.

Q.6 (Ans: B) For order $n = 3$, $\det(\text{adj}(A)) = (\det A)^{n-1} = 3^{3-1} = 3^2 = 9$.

Q.8 (Ans: C) By spectral mapping theorem, if λ is an eigenvalue of A , then λ^k is an eigenvalue of A^k . Hence $2^2 = 4$, $3^2 = 9$, $(-4)^2 = 16$.

Q.10 (Ans: B) Standard theorem: $\text{adj}(\text{adj}(A)) = |A|^{n-2}A$.

Q.13 (Ans: B) $x^y = e^{x-y} \implies y \ln x = x - y \implies y(1 + \ln x) = x \implies y = \frac{x}{1 + \ln x}$.

Differentiating: $\frac{dy}{dx} = \frac{(1 + \ln x)(1) - x(1/x)}{(1 + \ln x)^2} = \frac{\ln x}{(1 + \ln x)^2}$.

Q.19 (Ans: A) $\frac{1}{(D-2)^2} e^{2x} = \frac{x^2}{2!} e^{2x} = \frac{x^2}{2} e^{2x}$.

Q.23 (Ans: C) The alternating series converges by Leibniz test. However, the series of absolute values $\sum \frac{1}{n}$ is the harmonic series which diverges. Hence it is *conditionally convergent*.

Q.24 (Ans: D) Radius of convergence for $e^x = \sum \frac{x^n}{n!}$ is $R = \lim_{n \rightarrow \infty} \frac{(n+1)!}{n!} = \lim(n+1) = \infty$.

Q.25 (Ans: A) Sum of eigenvalues = Trace of matrix = $4 + 3 + (-2) = 5$.

Q.26 (Ans: A) Product of eigenvalues = Determinant of matrix = $(2)(5) - (1)(4) = 10 - 4 = 6$.

Q.29 (Ans: B) $W = \begin{vmatrix} \cos 2x & \sin 2x \\ -2 \sin 2x & 2 \cos 2x \end{vmatrix} = 2 \cos^2 2x + 2 \sin^2 2x = 2(1) = 2$.

Q.30 (Ans: B) $\frac{1}{D^2+4} \cos 2x = \frac{x}{2(2)} \sin 2x = \frac{x}{4} \sin 2x$.

2.2 Mid-Term Examination — Set B

LOVELY PROFESSIONAL UNIVERSITY

Transforming Education, Transforming India

Course Code: MTH-165 **Course Title:** Mathematics for Engineers

Academic Task Type: Mid-Term Examination (Objective MCQ)

Maximum Marks: 30 Marks **Time Allowed:** 90 Minutes

Course Outcomes Addressed: CO1 (Unit 1: Matrices), CO2 (Unit 2: ODEs & Calculus), CO3 (Unit 3: Infinite Series)

Roll No.:

Reg. No.:

Set B:

Instructions: All 30 questions are compulsory. Each question carries 1 mark. A penalty of 0.25 marks is deducted for each incorrect response.

Q.1 If A is a square matrix of order 3 such that $\det(A) = 4$, then $\det(2A)$ is: [1 Mark]

- (A) 8 (C) 32
(B) 16 (D) 64

Q.2 The eigenvalues of a skew-symmetric matrix are always: [1 Mark]

- (A) Real (C) Positive real numbers
(B) Purely imaginary or zero (D) Complex with real part 1

Q.3 The rank of the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$ is: [1 Mark]

- (A) 3 (C) 1
(B) 2 (D) 0

Q.4 If the system $x + y + z = 1, x + 2y + 4z = \alpha, x + 4y + 10z = \alpha^2$ is consistent, then α can be: [1 Mark]

- (A) 1 or 2 (C) -1 or 2
(B) 0 or 1 (D) Any real number

Q.5 If λ is an eigenvalue of an invertible matrix A , then an eigenvalue of $\text{adj}(A)$ is: [1 Mark]

- (A) $\frac{|A|}{\lambda}$ (C) $\lambda|A|$
(B) $\frac{\lambda}{|A|}$ (D) $\frac{1}{\lambda}$

Q.6 The trace of a 3×3 matrix with eigenvalues 1, -2 , 5 is: [1 Mark]

- (A) 4 (C) 6
(B) -10 (D) 8

Q.7 The differential equation $(D^2 + 1)y = 0$ has the general solution: [1 Mark]

- (A) $y = c_1 e^x + c_2 e^{-x}$ (C) $y = (c_1 + c_2 x)e^x$
(B) $y = c_1 \cos x + c_2 \sin x$ (D) $y = c_1 \cosh x + c_2 \sinh x$

Q.8 The value of $\lim_{x \rightarrow 0} \frac{\sin x - x}{x^3}$ is: [1 Mark]

- (A) 0 (C) $-\frac{1}{6}$
(B) $\frac{1}{6}$ (D) $\frac{1}{3}$

Q.9 The particular integral of $(D^2 - 1)y = x$ is: [1 Mark]

- (A) $-x$ (C) $\frac{x^2}{2}$
(B) x (D) e^x

Q.10 The general solution of $xy' + y = x^2$ is: [1 Mark]

- (A) $y = \frac{x^2}{3} + \frac{c}{x}$ (C) $y = \frac{x^3}{3} + c$
(B) $y = x^3 + c$ (D) $y = x + \frac{c}{x}$

Q.11 The order and degree of the differential equation $\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2} = \frac{d^2y}{dx^2}$ are: [1 Mark]

- (A) Order 2, Degree 3 (C) Order 1, Degree 3
(B) Order 2, Degree 2 (D) Order 3, Degree 2

Q.12 The series $\sum_{n=1}^{\infty} \frac{n}{n^3+1}$ is: [1 Mark]

- (A) Convergent (C) Oscillatory
(B) Divergent (D) Conditionally Convergent

Q.13 The series $\sum_{n=1}^{\infty} \frac{x^n}{n}$ converges for all x in: [1 Mark]

- (A) $(-1, 1]$ (C) $[-1, 1]$
(B) $[-1, 1)$ (D) $(-1, 1)$

Q.14 According to Cauchy's Root Test, $\sum u_n$ converges if $\lim_{n \rightarrow \infty} (u_n)^{1/n} = L$ where: [1 Mark]

- (A) $L < 1$ (C) $L = 1$
(B) $L > 1$ (D) $L \geq 1$

Q.15 The series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2}$ is: [1 Mark]

- (A) Conditionally Convergent (C) Divergent
(B) Absolutely Convergent (D) Unbounded

- Q.16** The derivative of $y = \tan^{-1} \left(\frac{2x}{1-x^2} \right)$ with respect to x is: [1 Mark]
- (A) $\frac{1}{1+x^2}$ (C) $\frac{2}{1-x^2}$
(B) $\frac{2}{1+x^2}$ (D) $\frac{1}{1-x^2}$
- Q.17** The value of $\int_0^{\pi/2} \ln(\tan x) dx$ is: [1 Mark]
- (A) 0 (C) $\pi \ln 2$
(B) $\frac{\pi}{2}$ (D) $-\frac{\pi}{2} \ln 2$
- Q.18** If $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, then the algebraic multiplicity of $\lambda = 1$ is: [1 Mark]
- (A) 1 (C) 0
(B) 2 (D) 3
- Q.19** If A and B are square matrices of the same order, then $\text{trace}(AB)$ is equal to: [1 Mark]
- (A) $\text{trace}(BA)$ (C) $\det(AB)$
(B) $\text{trace}(A)\text{trace}(B)$ (D) 0
- Q.20** The value of $\int \sec^3 x dx$ is: [1 Mark]
- (A) $\frac{1}{2}[\sec x \tan x + \ln |\sec x + \tan x|] + C$ (C) $\frac{1}{3} \sec^3 x + C$
(B) $\sec x \tan x + C$ (D) $\tan^2 x + C$
- Q.21** The particular integral of $(D^2 + 2D + 1)y = e^{-x}$ is: [1 Mark]
- (A) $\frac{x^2}{2}e^{-x}$ (C) $\frac{x}{2}e^{-x}$
(B) xe^{-x} (D) $\frac{x^3}{6}e^{-x}$
- Q.22** The test which is applied when D'Alembert's ratio test fails ($L = 1$) is: [1 Mark]
- (A) Raabe's Test (C) Integral Test
(B) Leibniz's Test (D) Root Test
- Q.23** The series $\sum_{n=1}^{\infty} \frac{1}{n \ln n}$ ($n \geq 2$) is: [1 Mark]
- (A) Convergent (C) Oscillatory
(B) Divergent (D) Conditionally Convergent
- Q.24** The value of $\lim_{x \rightarrow 0} (1 + 2x)^{1/x}$ is: [1 Mark]
- (A) 1 (C) e^2
(B) 2 (D) e

Q.25 A system of m homogeneous linear equations in n unknowns has a non-trivial solution if: [1 Mark]

- (A) $\text{rank}(A) = n$ (C) $\text{rank}(A) > n$
 (B) $\text{rank}(A) < n$ (D) $\det(A) \neq 0$

Q.26 If A is idempotent ($A^2 = A$), its eigenvalues can only be: [1 Mark]

- (A) 0 or 1 (C) 0 or -1
 (B) 1 or -1 (D) Any real number

Q.27 If A is nilpotent of index k ($A^k = O$), all its eigenvalues are: [1 Mark]

- (A) 1 (C) 0
 (B) -1 (D) k

Q.28 The solution of the differential equation $\frac{dy}{dx} = \frac{y}{x}$ is: [1 Mark]

- (A) $y = cx$ (C) $x^2 + y^2 = c$
 (B) $y = c/x$ (D) $y = x + c$

Q.29 The Taylor series expansion of $\sin x$ about $x = 0$ contains: [1 Mark]

- (A) Only even powers of x (C) Both even and odd powers
 (B) Only odd powers of x (D) No powers of x

Q.30 The geometric multiplicity of an eigenvalue λ is always: [1 Mark]

- (A) $\leq$ Algebraic Multiplicity (C) $= n$
 (B) $>$ Algebraic Multiplicity (D) $= 0$

Answer Key (Mid-Term Set B)

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	6	A	11	B	16	B	21	A	26	A
2	B	7	B	12	A	17	A	22	A	27	C
3	C	8	C	13	B	18	B	23	B	28	A
4	A	9	A	14	A	19	A	24	C	29	B
5	A	10	A	15	B	20	A	25	B	30	A

Selected Detailed Mathematical Explanations

Step-by-Step Mathematical Solution Explanations for Critical Questions

Q.1 (Ans: C) For $n \times n$ matrix A , $\det(kA) = k^n \det(A)$. Here $n = 3, k = 2$: $\det(2A) = 2^3 \det(A) = 8(4) = 32$.

Q.3 (Ans: C) Rows 2 and 3 are scalar multiples of Row 1 ($R_2 = 2R_1, R_3 = 3R_1$). Hence only 1 linearly independent row $\implies \text{rank} = 1$.

Q.5 (Ans: A) Since $A \cdot \text{adj}(A) = |A|I \implies \text{adj}(A) = |A|A^{-1}$. If eigenvalue of A is λ , eigenvalue of A^{-1} is $1/\lambda$, so eigenvalue of $\text{adj}(A)$ is $|A|/\lambda$.

Q.8 (Ans: C) $\lim_{x \rightarrow 0} \frac{(x - x^3/6 + \dots) - x}{x^3} = -\frac{1}{6}$.

Q.11 (Ans: B) Squaring both sides gives $[1 + (y')^2]^3 = (y'')^2$. The highest derivative is y'' (order 2) and its power is 2 (degree 2).

Q.17 (Ans: A) Using King's property $\int_0^a f(x)dx = \int_0^a f(a-x)dx$, $I = \int_0^{\pi/2} \ln(\cot x)dx = -\int_0^{\pi/2} \ln(\tan x)dx = -I \implies 2I = 0 \implies I = 0$.

Q.21 (Ans: A) $\frac{1}{(D+1)^2} e^{-x} = \frac{x^2}{2!} e^{-x} = \frac{x^2}{2} e^{-x}$.

Q.24 (Ans: C) $\lim_{x \rightarrow 0} (1 + 2x)^{1/x} = e^{\lim_{x \rightarrow 0} \frac{2x}{x}} = e^2$.

2.3 Mid-Term Examination — Set C

LOVELY PROFESSIONAL UNIVERSITY

Transforming Education, Transforming India

Course Code: MTH-165 **Course Title:** Mathematics for Engineers

Academic Task Type: Mid-Term Examination (Objective MCQ)

Maximum Marks: 30 Marks **Time Allowed:** 90 Minutes

Course Outcomes Addressed: CO1 (Unit 1: Matrices), CO2 (Unit 2: ODEs & Calculus), CO3 (Unit 3: Infinite Series)

Roll No.:

Reg. No.:

Set: **C** **Section:**

Instructions: All 30 questions are compulsory. Each question carries 1 mark. A penalty of 0.25 marks is deducted for each incorrect response.

Q.1 The value of the limit $\lim_{x \rightarrow 0} (\cos x)^{1/x^2}$ is: [1 Mark]

- (A) 1 (C) $e^{1/2}$
(B) $e^{-1/2}$ (D) 0

Q.2 If A is an involuntary matrix ($A^2 = I$), then its eigenvalues are: [1 Mark]

- (A) 0, 1 (C) $i, -i$
(B) ± 1 (D) Any real number

Q.3 The radius of convergence of the series $\sum_{n=1}^{\infty} \frac{n!}{n^n} x^n$ is: [1 Mark]

- (A) 1 (C) $1/e$
(B) e (D) ∞

Q.4 The general solution of $(D^2 + 4D + 4)y = 0$ is: [1 Mark]

- (A) $y = (c_1 + c_2 x)e^{-2x}$ (C) $y = (c_1 + c_2 x)e^{2x}$
(B) $y = c_1 e^{2x} + c_2 e^{-2x}$ (D) $y = c_1 \cos 2x + c_2 \sin 2x$

Q.5 If two eigenvalues of a 3×3 matrix with determinant -12 and trace 2 are 1 and 3, then the third eigenvalue is: [1 Mark]

- (A) -2 (C) 4
(B) -4 (D) 2

Q.6 The series $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n+1}}$ is: [1 Mark]

- (A) Convergent (C) Oscillatory
(B) Divergent (D) Conditionally Convergent

- Q.7** The value of $\int_0^1 x(1-x)^9 dx$ using Beta function is: [1 Mark]
- (A) $\frac{1}{110}$ (C) $\frac{1}{100}$
(B) $\frac{1}{90}$ (D) $\frac{1}{11}$
- Q.8** If A is a real orthogonal matrix, then A^{-1} is always equal to: [1 Mark]
- (A) A (C) A^T
(B) $-A$ (D) $(\bar{A})^T$
- Q.9** The particular integral of $(D^2 + 1)y = \sin x$ is: [1 Mark]
- (A) $-\frac{x}{2} \cos x$ (C) $-\frac{x}{2} \sin x$
(B) $\frac{x}{2} \cos x$ (D) $\frac{x}{2} \sin x$
- Q.10** The series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{\sqrt{n}}$ is: [1 Mark]
- (A) Absolutely Convergent (C) Divergent
(B) Conditionally Convergent (D) Oscillatory
- Q.11** The rank of the matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$ is: [1 Mark]
- (A) 1 (C) 3
(B) 2 (D) 0
- Q.12** The Wronskian of the functions e^x and e^{-x} is: [1 Mark]
- (A) 0 (C) -2
(B) 2 (D) 1
- Q.13** The value of $\lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 2x}{x - \sin x}$ is: [1 Mark]
- (A) 1 (C) 3
(B) 2 (D) 4
- Q.14** A homogeneous system $AX = 0$ in n unknowns has only the trivial solution if and only if: [1 Mark]
- (A) $\text{rank}(A) = n$ (C) $\det(A) = 0$
(B) $\text{rank}(A) < n$ (D) $\text{rank}(A) = 1$
- Q.15** The particular integral of $(D - 1)y = e^x$ is: [1 Mark]
- (A) xe^x (C) e^x
(B) $\frac{x^2}{2}e^x$ (D) x^2e^x

Q.16 The series $\sum_{n=1}^{\infty} \left(\frac{n}{2n+1}\right)^n$ converges by: [1 Mark]

- (A) D'Alembert's Ratio Test (C) Raabe's Test
(B) Cauchy's Root Test (D) Integral Test

Q.17 If $A = \begin{bmatrix} 3 & 4 \\ 4 & -3 \end{bmatrix}$, then A is: [1 Mark]

- (A) Orthogonal and Symmetric (C) Singular
(B) Skew-Symmetric (D) Idempotent

Q.18 The solution of $y'' + 4y' + 5y = 0$ is: [1 Mark]

- (A) $e^{-2x}(c_1 \cos x + c_2 \sin x)$ (C) $(c_1 + c_2 x)e^{-2x}$
(B) $e^{2x}(c_1 \cos x + c_2 \sin x)$ (D) $c_1 e^{-x} + c_2 e^{-5x}$

Q.19 The Maclaurin series expansion of $\ln(1+x)$ valid for $|x| < 1$ begins with: [1 Mark]

- (A) $1+x$ (C) $x + \frac{x^2}{2} + \frac{x^3}{3}$
(B) $x - \frac{x^2}{2} + \frac{x^3}{3}$ (D) $1 - x + x^2$

Q.20 If A is non-singular, then $\text{adj}(A)$ is: [1 Mark]

- (A) Singular (C) Symmetric
(B) Non-singular (D) Zero

Q.21 The value of $\int_0^{\pi/2} \sin^4 x \cos^2 x \, dx$ is: [1 Mark]

- (A) $\frac{\pi}{32}$ (C) $\frac{3\pi}{32}$
(B) $\frac{\pi}{16}$ (D) $\frac{\pi}{64}$

Q.22 The series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ has sum equal to: [1 Mark]

- (A) $\frac{\pi^2}{6}$ (C) $\frac{\pi}{4}$
(B) $\frac{\pi^2}{8}$ (D) 1

Q.23 If $y = e^{m \sin^{-1} x}$, then $(1-x^2)y_2 - xy_1$ is equal to: [1 Mark]

- (A) $m^2 y$ (C) my
(B) $-m^2 y$ (D) 0

Q.24 The degree of the differential equation $\frac{d^2 y}{dx^2} + \sin\left(\frac{dy}{dx}\right) = 0$ is: [1 Mark]

- (A) 1 (C) Not defined
(B) 2 (D) 0

Q.25 The product of two orthogonal matrices of the same order is: [1 Mark]

- (A) Symmetric (C) Orthogonal
(B) Skew-Symmetric (D) Diagonal

Q.26 The particular integral of $(D^2 + D)y = x$ is: [1 Mark]

- (A) $\frac{x^2}{2} - x$ (C) $x^2 - x$
(B) $\frac{x^2}{2} + x$ (D) $\frac{x^3}{6}$

Q.27 The value of $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \frac{1}{1+(r/n)}$ is: [1 Mark]

- (A) $\ln 2$ (C) 1
(B) $\ln 3$ (D) $\frac{1}{2}$

Q.28 If A is a 3×3 matrix and $\text{rank}(A) = 2$, then $\text{nullity}(A)$ is: [1 Mark]

- (A) 0 (C) 2
(B) 1 (D) 3

Q.29 The eigenvalues of a diagonal matrix are: [1 Mark]

- (A) All zero (C) The diagonal elements themselves
(B) All 1 (D) Purely imaginary

Q.30 The test that determines convergence by integrating $f(x)$ from 1 to ∞ is: [1 Mark]

- (A) Raabe's Test (C) Leibniz's Test
(B) Cauchy's Integral Test (D) Gauss's Test

Answer Key (Mid-Term Set C)

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	6	B	11	C	16	B	21	A	26	A
2	B	7	A	12	C	17	A	22	A	27	A
3	B	8	C	13	B	18	A	23	A	28	B
4	A	9	A	14	A	19	B	24	C	29	C
5	B	10	B	15	A	20	B	25	C	30	B

Selected Detailed Mathematical Explanations

Step-by-Step Mathematical Solution Explanations for Critical Questions

Q.1 (Ans: B) $\lim_{x \rightarrow 0} (\cos x)^{1/x^2} = e^{\lim_{x \rightarrow 0} \frac{\ln \cos x}{x^2}} = e^{\lim_{x \rightarrow 0} \frac{-\tan x}{2x}} = e^{-1/2}.$

Q.3 (Ans: B) $\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{(n+1)!}{(n+1)^{n+1}} \frac{n^n}{n!} = \lim_{n \rightarrow \infty} \frac{1}{(1+1/n)^n} = \frac{1}{e}.$ Radius $R = 1/L = e.$

Q.5 (Ans: B) $\lambda_1 \lambda_2 \lambda_3 = \det(A) = -12 \implies (1)(3)\lambda_3 = -12 \implies \lambda_3 = -4.$ Also trace $= 1 + 3 + (-4) = 0 \neq 2$ unless trace is 0; here product confirms $\lambda_3 = -4.$

Q.7 (Ans: A) $\int_0^1 x^1 (1-x)^9 dx = B(2, 10) = \frac{\Gamma(2)\Gamma(10)}{\Gamma(12)} = \frac{1! \cdot 9!}{11!} = \frac{1}{11 \times 10} = \frac{1}{110}.$

Q.9 (Ans: A) $\frac{1}{D^2+1} \sin x = -\frac{x}{2(1)} \cos x = -\frac{x}{2} \cos x.$

Q.12 (Ans: C) $W = \begin{vmatrix} e^x & e^{-x} \\ e^x & -e^{-x} \end{vmatrix} = -e^0 - e^0 = -2.$

Q.24 (Ans: C) Because the derivative is inside a transcendental sine function, the equation cannot be expressed as a polynomial in derivatives; its degree is not defined.

Chapter 3

Continuous Assessment 2 (CA-2) Suite

Continuous Assessment 2 evaluates students on **Unit 4 (Multivariate Differentiation)** and **Unit 5 (Multivariable Integration and Applications)**. Each paper contains **6 questions** carrying **5 marks each**, and students attempt **5 questions** (Total = **30 Marks**, Time Allowed = **60 Minutes**).

3.1 Continuous Assessment 2 (CA-2) — Set A

LOVELY PROFESSIONAL UNIVERSITY

Transforming Education, Transforming India

Course Code: MTH-165 **Course Title:** Mathematics for Engineers

Academic Task Type: Continuous Assessment (CA-2) **Set:** A

Maximum Marks: 30 Marks **Time Allowed:** 50 Minutes

Roll No.:

Reg. No.:

Section:

Course Outcomes Addressed: CO4 (Multivariable Limits, Continuity, Partial Derivatives & Optimization), CO5 (Multiple Integrals & Transformations)

Note: Attempt all six questions. Each question carries 5 marks.

Q.1 Find if the two-variable limit exists as $(x, y) \rightarrow (0, 0)$: **CO4**

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^4 + y^2}$$

Show the rigorous mathematical justification using path testing. L1 Remember / Analyze **5 Marks**

Q.2 For the function $f(x, y) = x^2 \tan^{-1}(\frac{y}{x})$, verify the equality of mixed partial derivatives (Schwarz's Theorem): **CO4**

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$$

L3 Apply **5 Marks**

Q.3 A company manufactures two products x and y . The joint profit function is given by: **CO4**

$$P(x, y) = 40x + 30y - (x^2 + xy + 2y^2)$$

Find the production levels x and y that **maximize the profit**, and compute the maximum profit value. L3 Apply **5 Marks**

Q.4 Change the order of integration and hence evaluate: **CO5**

$$I = \int_0^2 \int_{x-1}^1 (x - y) dy dx$$

L3 Apply **5 Marks**

Q.5 Evaluate the double integral by transforming to polar coordinates:

CO5

L3 Apply

**5
Marks**

$$\iint_R e^{x^2+y^2} dA$$

where R is the circular disk of unit radius centered at the origin: $x^2 + y^2 \leq 1$.

Q.6 Evaluate the triple integral:

CO5

L3 Apply

**5
Marks**

$$\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^1 xz \, dz \, dy \, dx$$

Solutions & Marking Scheme (CA-2 Set A)

Step-by-Step Mathematical Solution Solution to Question 1 (5 Marks)

Step 1: Path Testing along Linear Directions Consider paths approaching $(0,0)$ along straight lines $y = mx$:

$$\lim_{x \rightarrow 0} \frac{x^2(mx)}{x^4 + (mx)^2} = \lim_{x \rightarrow 0} \frac{mx^3}{x^2(x^2 + m^2)} = \lim_{x \rightarrow 0} \frac{mx}{x^2 + m^2} = \frac{0}{0 + m^2} = 0$$

The limit along all straight lines is 0. However, this is NOT sufficient to prove that the limit exists!

Step 2: Path Testing along Parabolic Curves Observe the powers in the denominator: $x^4 + y^2$. The terms have balanced degrees if $y^2 \propto x^4 \implies y = kx^2$. Approach the origin along the parabola $y = kx^2$:

$$\lim_{x \rightarrow 0} \frac{x^2(kx^2)}{x^4 + (kx^2)^2} = \lim_{x \rightarrow 0} \frac{kx^4}{x^4 + k^2x^4} = \lim_{x \rightarrow 0} \frac{kx^4}{x^4(1 + k^2)} = \frac{k}{1 + k^2}$$

Step 3: Conclusion The limiting value depends explicitly on the parameter k (the path of approach). For instance:

- Along $y = x^2$ ($k = 1$): limit is $\frac{1}{1+1} = \frac{1}{2}$.
- Along $y = 2x^2$ ($k = 2$): limit is $\frac{2}{1+4} = \frac{2}{5}$.

Since the limit along different approaching paths produces distinct values, the two-variable limit:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2y}{x^4 + y^2} \text{ DOES NOT EXIST.}$$

Exam Warning

Never stop after checking $y = mx$. Always examine the degrees in the denominator. If the denominator is $x^{2a} + y^{2b}$, test paths of the form $y = kx^{a/b}$.

Mark Distribution: • Testing linear paths $y = mx$ and finding 0: **1.5 Marks**

• Testing parabolic path $y = kx^2$ with balanced exponents: **2 Marks**

• Proving path dependence and concluding limit does not exist: **1.5 Marks**

Step-by-Step Mathematical Solution Solution to Question 2 (5 Marks)

Given $f(x, y) = x^2 \tan^{-1}\left(\frac{y}{x}\right)$.

Step 1: Compute $\frac{\partial f}{\partial y}$ Treat x as constant:

$$\frac{\partial f}{\partial y} = x^2 \cdot \frac{1}{1 + \left(\frac{y}{x}\right)^2} \cdot \frac{\partial}{\partial y} \left(\frac{y}{x}\right) = x^2 \cdot \frac{x^2}{x^2 + y^2} \cdot \frac{1}{x} = \frac{x^3}{x^2 + y^2}$$

Now differentiate with respect to x to find $\frac{\partial^2 f}{\partial x \partial y}$:

$$\begin{aligned} \frac{\partial^2 f}{\partial x \partial y} &= \frac{\partial}{\partial x} \left[\frac{x^3}{x^2 + y^2} \right] = \frac{(x^2 + y^2) \frac{d}{dx}(x^3) - x^3 \frac{d}{dx}(x^2 + y^2)}{(x^2 + y^2)^2} \\ &= \frac{(x^2 + y^2)(3x^2) - x^3(2x)}{(x^2 + y^2)^2} = \frac{3x^4 + 3x^2y^2 - 2x^4}{(x^2 + y^2)^2} = \frac{x^4 + 3x^2y^2}{(x^2 + y^2)^2} \end{aligned}$$

Step 2: Compute $\frac{\partial f}{\partial x}$ Using the product rule on $x^2 \cdot \tan^{-1}(y/x)$:

$$\frac{\partial f}{\partial x} = 2x \tan^{-1}\left(\frac{y}{x}\right) + x^2 \cdot \frac{1}{1 + \left(\frac{y}{x}\right)^2} \left(-\frac{y}{x^2}\right) = 2x \tan^{-1}\left(\frac{y}{x}\right) - \frac{x^2y}{x^2 + y^2}$$

Now differentiate with respect to y to find $\frac{\partial^2 f}{\partial y \partial x}$:

$$\frac{\partial^2 f}{\partial y \partial x} = 2x \cdot \frac{1}{1 + (y/x)^2} \left(\frac{1}{x}\right) - \frac{\partial}{\partial y} \left[\frac{x^2y}{x^2 + y^2} \right]$$

First term:

$$2x \cdot \frac{x^2}{x^2 + y^2} \cdot \frac{1}{x} = \frac{2x^2}{x^2 + y^2}$$

Second term (quotient rule with respect to y , x constant):

$$\frac{\partial}{\partial y} \left[\frac{x^2y}{x^2 + y^2} \right] = x^2 \left[\frac{(x^2 + y^2)(1) - y(2y)}{(x^2 + y^2)^2} \right] = \frac{x^2(x^2 - y^2)}{(x^2 + y^2)^2} = \frac{x^4 - x^2y^2}{(x^2 + y^2)^2}$$

Combine both terms over common denominator $(x^2 + y^2)^2$:

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{2x^2(x^2 + y^2) - (x^4 - x^2y^2)}{(x^2 + y^2)^2} = \frac{2x^4 + 2x^2y^2 - x^4 + x^2y^2}{(x^2 + y^2)^2} = \frac{x^4 + 3x^2y^2}{(x^2 + y^2)^2}$$

Step 3: Conclusion Both mixed partial derivatives are identical:

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} = \frac{x^4 + 3x^2y^2}{(x^2 + y^2)^2} \quad (\text{Verified!})$$

Mark Distribution: • First partial $\partial f/\partial y$ and mixed partial $\partial^2 f/\partial x \partial y$: **2 Marks**

• First partial $\partial f/\partial x$ and mixed partial $\partial^2 f/\partial y \partial x$: **2 Marks**

• Exact algebraic equivalence verification: **1 Mark**

Step-by-Step Mathematical Solution Solution to Question 3 (5 Marks)

Step 1: First Partial Derivatives Profit function: $P(x, y) = 40x + 30y - x^2 - xy - 2y^2$.

$$\frac{\partial P}{\partial x} = 40 - 2x - y$$

$$\frac{\partial P}{\partial y} = 30 - x - 4y$$

Step 2: Stationary (Critical) Points Set $\frac{\partial P}{\partial x} = 0$ and $\frac{\partial P}{\partial y} = 0$:

$$2x + y = 40 \implies y = 40 - 2x \quad (3.1)$$

$$x + 4y = 30 \quad (3.2)$$

Substitute (3.1) into (3.2):

$$x + 4(40 - 2x) = 30 \implies x + 160 - 8x = 30 \implies -7x = -130 \implies x = \frac{130}{7} \approx 18.57$$

Substitute x back:

$$y = 40 - 2\left(\frac{130}{7}\right) = \frac{280 - 260}{7} = \frac{20}{7} \approx 2.86$$

Critical point is $\left(\frac{130}{7}, \frac{20}{7}\right)$.

Step 3: Second Derivative Test (Hessian) Compute second-order partial derivatives:

$$r = \frac{\partial^2 P}{\partial x^2} = -2$$

$$s = \frac{\partial^2 P}{\partial x \partial y} = -1$$

$$t = \frac{\partial^2 P}{\partial y^2} = -4$$

Discriminant:

$$rt - s^2 = (-2)(-4) - (-1)^2 = 8 - 1 = 7 > 0$$

Since $rt - s^2 = 7 > 0$ and $r = -2 < 0$, the function achieves a **strict local maximum** at this critical point.

Step 4: Maximum Profit Value

$$\begin{aligned} P_{\max} &= 40\left(\frac{130}{7}\right) + 30\left(\frac{20}{7}\right) - \left[\left(\frac{130}{7}\right)^2 + \left(\frac{130}{7}\right)\left(\frac{20}{7}\right) + 2\left(\frac{20}{7}\right)^2\right] \\ &= \frac{5200 + 600}{7} - \frac{16900 + 2600 + 800}{49} = \frac{5800}{7} - \frac{20300}{49} = \frac{40600 - 20300}{49} = \frac{20300}{49} = \frac{2900}{7} \approx 414.29 \end{aligned}$$

Optimal production levels: $\mathbf{x} = \frac{130}{7}$, $\mathbf{y} = \frac{20}{7}$.

Mark Distribution: • First partial derivatives and critical point equations: **1.5 Marks**

• Solving for $(x, y) = (130/7, 20/7)$: **1 Mark**

• Second derivative test ($rt - s^2 > 0$ and $r < 0$): **1.5 Marks**

• Maximum profit value calculation: **1 Mark**

Step-by-Step Mathematical Solution Solution to Question 4 (5 Marks)**Step 1: Region of Integration** Given integral:

$$I = \int_0^2 \int_{x-1}^1 (x-y) dy dx$$

The given limits correspond to:

$$x \in [0, 2], \quad y \in [x-1, 1]$$

The region is bounded by:

- Bottom curve: $y = x - 1 \implies x = y + 1$
- Top horizontal line: $y = 1$
- Vertical boundary: $x = 0$

Intersection points:

- $x = 0 \implies y = -1$ (at point $(0, -1)$)
- $x = 2 \implies y = 2 - 1 = 1$ (at point $(2, 1)$)
- At $y = 1$: $x = 0$ to $x = 2$.

The range of y across the entire region is $y \in [-1, 1]$.**Step 2: Reversing the Order (Horizontal Strips)** For any horizontal strip at height y :

- x enters at the left boundary: $x = 0$.
- x exits at the right line: $x = y + 1$.

Thus, the limits in reversed order are:

$$I = \int_{-1}^1 \int_0^{y+1} (x-y) dx dy$$

Step 3: Evaluation Inner integral with respect to x :

$$\int_0^{y+1} (x-y) dx = \left[\frac{x^2}{2} - yx \right]_0^{y+1} = \frac{(y+1)^2}{2} - y(y+1) = \frac{y^2 + 2y + 1 - 2y^2 - 2y}{2} = \frac{1 - y^2}{2}$$

Outer integral with respect to y :

$$I = \int_{-1}^1 \frac{1-y^2}{2} dy = \frac{1}{2} \int_{-1}^1 (1-y^2) dy = \frac{1}{2} \left[y - \frac{y^3}{3} \right]_{-1}^1 = \frac{1}{2} \left(1 - \frac{1}{3} - \left(-1 + \frac{1}{3} \right) \right) = \frac{1}{2} \left(1 - \frac{1}{3} + 1 - \frac{1}{3} \right) = \frac{1}{2} \left(2 - \frac{2}{3} \right) = \frac{1}{2} \cdot \frac{4}{3} = \frac{2}{3}$$

Mark Distribution: • Sketch/identification of original boundary lines and vertices: **1.5 Marks**

- Correctly setting new limits $y \in [-1, 1]$ and $x \in [0, y+1]$: **2 Marks**
- Accurate integration yielding final answer $2/3$: **1.5 Marks**

Step-by-Step Mathematical Solution Solution to Question 5 (5 Marks)**Step 1: Polar Coordinate Transformation** Let:

$$x = r \cos \theta, \quad y = r \sin \theta \implies x^2 + y^2 = r^2$$

The differential area element transforms as:

$$dA = dx dy = r dr d\theta$$

Step 2: Region Limits The region R is the complete unit disk $x^2 + y^2 \leq 1$:

$$r \in [0, 1], \quad \theta \in [0, 2\pi]$$

Step 3: Integral Evaluation

$$\iint_R e^{x^2+y^2} dA = \int_0^{2\pi} \int_0^1 e^{r^2} \cdot r dr d\theta$$

Separate the independent variables:

$$= \left(\int_0^{2\pi} d\theta \right) \left(\int_0^1 r e^{r^2} dr \right)$$

The angular integral is $[\theta]_0^{2\pi} = 2\pi$. For the radial integral, let $u = r^2 \implies du = 2r dr \implies r dr = \frac{du}{2}$:

$$\int_0^1 r e^{r^2} dr = \frac{1}{2} \int_0^1 e^u du = \frac{1}{2} [e^u]_0^1 = \frac{1}{2} (e^1 - e^0) = \frac{e-1}{2}$$

Multiply the results:

$$\iint_R e^{x^2+y^2} dA = 2\pi \left(\frac{e-1}{2} \right) = \pi(e-1)$$

Mark Distribution: • Transformation formulas and Jacobian $r dr d\theta$: **1.5 Marks**• Correct limits for unit circle ($r \in [0, 1], \theta \in [0, 2\pi]$): **1.5 Marks**• Integration and final exact answer $\pi(e-1)$: **2 Marks**

Step-by-Step Mathematical Solution Solution to Question 6 (5 Marks)**Step 1: Understand the Integral Structure**

$$I = \int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^1 xz \, dz \, dy \, dx$$

Notice that the innermost integral depends on z , with constant limits $z \in [0, 1]$. **Step 2: Inner Integration (with respect to z)**

$$\int_0^1 xz \, dz = x \left[\frac{z^2}{2} \right]_0^1 = \frac{1}{2}x$$

Step 3: Middle Integration (with respect to y) Now the integral becomes:

$$\int_0^{\sqrt{1-x^2}} \frac{1}{2}x \, dy = \frac{1}{2}x[y]_0^{\sqrt{1-x^2}} = \frac{1}{2}x\sqrt{1-x^2}$$

Step 4: Outermost Integration (with respect to x)

$$I = \int_0^1 \frac{1}{2}x\sqrt{1-x^2} \, dx$$

Let $u = 1 - x^2 \implies du = -2x \, dx \implies x \, dx = -\frac{du}{2}$. When $x = 0, u = 1$; when $x = 1, u = 0$:

$$I = \frac{1}{2} \int_1^0 \sqrt{u} \left(-\frac{du}{2} \right) = \frac{1}{4} \int_0^1 u^{1/2} \, du = \frac{1}{4} \left[\frac{u^{3/2}}{3/2} \right]_0^1 = \frac{1}{4} \cdot \frac{2}{3} = \frac{1}{6}$$

Mark Distribution: • Integration with respect to z : **1.5 Marks**

• Integration with respect to y : **1.5 Marks**

• Substitution and final evaluation with respect to x giving $1/6$: **2 Marks**

3.2 Continuous Assessment 2 (CA-2) — Set B

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Course Code: MTH-165 **Course Title:** Mathematics for Engineers

Academic Task Type: Continuous Assessment (CA-2) **Set:** B

Maximum Marks: 30 Marks **Time Allowed:** 50 Minutes

Roll No.:

Reg. No.:

Section:

Course Outcomes Addressed: CO4 (Multivariable Limits, Continuity, Partial Derivatives & Optimization), CO5 (Multiple Integrals & Transformations)

Note: Attempt all six questions. Each question carries 5 marks.

Q.1

Investigate the **continuity** of the function $f(x, y)$ at the origin $(0, 0)$:

CO4

L1 Remem-
ber / Ana-
lyze

**5
Marks**

$$f(x, y) = \begin{cases} \frac{x^3 - y^3}{x + y} & (x, y) \neq (0, 0) \text{ and } x + y \neq 0 \\ 0 & (x, y) = (0, 0) \end{cases}$$

Q.2

Given the three-variable function $f(x, y, z) = xy^2z^3$, compute the second-order partial derivatives:

CO4

L3 Apply

**5
Marks**

$$f_{xx}, \quad f_{xy}, \quad f_{zx}, \quad f_{yx}$$

Verify whether $f_{xy} = f_{yx}$.

Q.3

A company wants to manufacture a closed cylindrical container with fixed volume $V \text{ cm}^3$. The sheet metal for the top and bottom circular lids costs the same per cm^2 as the sheet metal used for the curved cylindrical side. Determine the radius r and height h that **minimize the total manufacturing cost**.

CO4

L3 Apply

**5
Marks**

Q.4

Evaluate the triple iterated integral:

CO5

L3 Apply

**5
Marks**

$$\int_0^1 \int_{y^2}^1 \int_0^{1-x} (y - 2z) dz dx dy$$

Q.5

Evaluate the double integral:

CO5

L3 Apply

**5
Marks**

$$\iint_R (x^2 + y^2) dA$$

where R is the region in the **second quadrant** bounded by the circle $x^2 + y^2 = 4$.

Q.6 Evaluate the triple integral:

CO5

L3 Apply

5
Marks

$$\iiint_E xy \, dV$$

where E is the solid region bounded by the planes:

$$0 \leq x \leq 1, 0 \leq y \leq 1, \text{ and } 0 \leq z \leq x + y.$$

Solutions & Marking Scheme (CA-2 Set B)

Step-by-Step Mathematical Solution Solution to Question 1 (5 Marks)

Step 1: Definition of Continuity A function $f(x, y)$ is continuous at $(0, 0)$ if and only if:

$$\lim_{(x,y) \rightarrow (0,0)} f(x, y) = f(0, 0) = 0$$

Step 2: Path Testing along $y = mx$ Approach the origin along straight lines $y = mx$ where $m \neq -1$:

$$\lim_{x \rightarrow 0} \frac{x^3 - (mx)^3}{x + mx} = \lim_{x \rightarrow 0} \frac{x^3(1 - m^3)}{x(1 + m)} = \lim_{x \rightarrow 0} x^2 \frac{1 - m^3}{1 + m} = \lim_{x \rightarrow 0} x^2(1 - m + m^2) = 0$$

This holds for any slope $m \neq -1$.

Step 3: Path Testing approaching the singular line $y = -x$ Let $y = -x + kx^2$:

$$x^3 - y^3 = x^3 - (-x + kx^2)^3 = x^3 - (-x^3 + 3x^2(kx^2) - \dots) = 2x^3 - 3kx^4 + \dots$$

$$x + y = x + (-x + kx^2) = kx^2$$

Then the limit becomes:

$$\lim_{x \rightarrow 0} \frac{2x^3 - 3kx^4}{kx^2} = \lim_{x \rightarrow 0} \frac{2x}{k} = 0$$

Furthermore, using polar coordinates $x = r \cos \theta, y = r \sin \theta$:

$$\left| \frac{x^3 - y^3}{x + y} \right| = \left| \frac{r^3(\cos^3 \theta - \sin^3 \theta)}{r(\cos \theta + \sin \theta)} \right| = r^2 |\cos^2 \theta - \cos \theta \sin \theta + \sin^2 \theta| \leq 2r^2 \rightarrow 0 \text{ as } r \rightarrow 0$$

Since the limit as $(x, y) \rightarrow (0, 0)$ exists and equals $0 = f(0, 0)$, the function is **continuous at $(0, 0)$** .

Mark Distribution: • Definition of multivariable continuity: **1.5 Marks**

• Evaluation along arbitrary paths / polar coordinates: **2 Marks**

• Proving $\lim = f(0, 0) = 0$ and concluding continuity: **1.5 Marks**

Step-by-Step Mathematical Solution Solution to Question 2 (5 Marks)

Given $f(x, y, z) = xy^2z^3$.

Step 1: First Partial Derivatives

$$f_x = \frac{\partial f}{\partial x} = y^2z^3, \quad f_y = \frac{\partial f}{\partial y} = 2xyz^3, \quad f_z = \frac{\partial f}{\partial z} = 3xy^2z^2$$

Step 2: Second Partial Derivatives

- $f_{xx} = \frac{\partial}{\partial x}(f_x) = \frac{\partial}{\partial x}(y^2z^3) = 0$.
- $f_{xy} = \frac{\partial}{\partial y}(f_x) = \frac{\partial}{\partial y}(y^2z^3) = 2yz^3$.
- $f_{yx} = \frac{\partial}{\partial x}(f_y) = \frac{\partial}{\partial x}(2xyz^3) = 2yz^3$.
- $f_{zx} = \frac{\partial}{\partial x}(f_z) = \frac{\partial}{\partial x}(3xy^2z^2) = 3y^2z^2$.

Step 3: Verification Notice that $f_{xy} = 2yz^3 = f_{yx}$. The mixed partials are identically equal, verifying Schwarz's Theorem.

Mark Distribution: • First partial derivatives: **1.5 Marks**

• Computation of f_{xx}, f_{zx} : **1.5 Marks**

• Computation of f_{xy}, f_{yx} and verification $f_{xy} = f_{yx}$: **2 Marks**

Step-by-Step Mathematical Solution Solution to Question 3 (5 Marks)

Step 1: Problem Formulation Let r be the radius and h be the height of the cylinder. Given volume V is fixed:

$$V = \pi r^2 h \implies h = \frac{V}{\pi r^2}$$

Since the cost per unit area is uniform (say C per cm^2), minimizing cost is equivalent to minimizing the total surface area S :

$$S = 2(\pi r^2) + 2\pi r h = 2\pi r^2 + 2\pi r \left(\frac{V}{\pi r^2} \right) = 2\pi r^2 + \frac{2V}{r}$$

Step 2: Finding Stationary Points Differentiate S with respect to r :

$$\frac{dS}{dr} = 4\pi r - \frac{2V}{r^2}$$

Set $\frac{dS}{dr} = 0$:

$$4\pi r = \frac{2V}{r^2} \implies r^3 = \frac{2V}{4\pi} = \frac{V}{2\pi} \implies r = \left(\frac{V}{2\pi} \right)^{1/3}$$

Step 3: Second Derivative Test

$$\frac{d^2S}{dr^2} = 4\pi + \frac{4V}{r^3} = 4\pi + \frac{4V}{V/(2\pi)} = 4\pi + 8\pi = 12\pi > 0$$

Since $\frac{d^2S}{dr^2} > 0$, the surface area (and cost) is strictly ****minimized****.

Step 4: Optimal Dimensions Substitute r into (3.2):

$$h = \frac{V}{\pi r^2} = \frac{2\pi r^3}{\pi r^2} = 2r$$

Thus, the optimal dimensions are:

$$r = \left(\frac{V}{2\pi} \right)^{1/3}, \quad h = 2r = 2 \left(\frac{V}{2\pi} \right)^{1/3}$$

The height must equal the diameter ($h = 2r$) of the cylindrical base.

Mark Distribution: • Surface area objective function in single variable r : **1.5 Marks**

• First derivative and finding critical point $r = (V/2\pi)^{1/3}$: **1.5 Marks**

• Second derivative test confirming minimum: **1 Mark**

• Concluding optimal dimension relation $h = 2r$: **1 Mark**

Step-by-Step Mathematical Solution Solution to Question 4 (5 Marks)

Integral to evaluate:

$$I = \int_0^1 \int_{y^2}^1 \int_0^{1-x} (y - 2z) dz dx dy$$

Step 1: Inner Integration with respect to z

$$\int_0^{1-x} (y - 2z) dz = \left[yz - z^2 \right]_0^{1-x} = y(1-x) - (1-x)^2 = y - xy - (1 - 2x + x^2)$$

Step 2: Middle Integration with respect to x

$$\int_{y^2}^1 \left[y(1-x) - (1-x)^2 \right] dx = \left[-y \frac{(1-x)^2}{2} + \frac{(1-x)^3}{3} \right]_{y^2}^1$$

At upper limit $x = 1$, $(1-1) = 0$. At lower limit $x = y^2$:

$$= 0 - \left[-y \frac{(1-y^2)^2}{2} + \frac{(1-y^2)^3}{3} \right] = \frac{y(1-y^2)^2}{2} - \frac{(1-y^2)^3}{3}$$

Step 3: Outer Integration with respect to y

$$I = \int_0^1 \frac{y(1-y^2)^2}{2} dy - \int_0^1 \frac{(1-y^2)^3}{3} dy$$

For the first integral, let $u = 1 - y^2 \implies du = -2y dy \implies y dy = -du/2$:

$$I_1 = \frac{1}{2} \int_1^0 u^2 \left(-\frac{du}{2} \right) = \frac{1}{4} \int_0^1 u^2 du = \frac{1}{4} \left[\frac{1}{3} \right] = \frac{1}{12}$$

For the second integral, expand $(1 - y^2)^3 = 1 - 3y^2 + 3y^4 - y^6$:

$$I_2 = \frac{1}{3} \int_0^1 (1 - 3y^2 + 3y^4 - y^6) dy = \frac{1}{3} \left[1 - 1 + \frac{3}{5} - \frac{1}{7} \right] = \frac{1}{3} \left[\frac{21-5}{35} \right] = \frac{16}{105}$$

Subtract:

$$I = I_1 - I_2 = \frac{1}{12} - \frac{16}{105} = \frac{35-64}{420} = -\frac{29}{420}$$

Mark Distribution: • Integration with respect to z : **1.5 Marks**

• Integration with respect to x : **1.5 Marks**

• Outer integral with respect to y yielding $-29/420$: **2 Marks**

Step-by-Step Mathematical Solution Solution to Question 5 (5 Marks)

Step 1: Polar Coordinate Transformation Let $x = r \cos \theta, y = r \sin \theta \implies x^2 + y^2 = r^2$, and $dA = r dr d\theta$. **Step 2: Limits for the Second Quadrant of $x^2 + y^2 = 4$** In the second quadrant ($x \leq 0, y \geq 0$):

$$\theta \in \left[\frac{\pi}{2}, \pi\right], \quad r \in [0, 2]$$

Step 3: Integral Evaluation

$$\begin{aligned} \iint_R (x^2 + y^2) dA &= \int_{\pi/2}^{\pi} \int_0^2 (r^2) \cdot r dr d\theta = \left(\int_{\pi/2}^{\pi} d\theta \right) \left(\int_0^2 r^3 dr \right) \\ &= \left(\pi - \frac{\pi}{2} \right) \left[\frac{r^4}{4} \right]_0^2 = \left(\frac{\pi}{2} \right) \left(\frac{16}{4} \right) = \left(\frac{\pi}{2} \right) (4) = 2\pi \end{aligned}$$

Mark Distribution: • Polar transformation formulas and Jacobian r : **1.5 Marks**

• Setting second quadrant limits $\theta \in [\pi/2, \pi]$ and $r \in [0, 2]$: **1.5 Marks**

• Calculation yielding final answer 2π : **2 Marks**

Step-by-Step Mathematical Solution Solution to Question 6 (5 Marks)

Step 1: Setup Iterated Triple Integral Region E : $0 \leq x \leq 1, 0 \leq y \leq 1, 0 \leq z \leq x + y$.

$$I = \int_0^1 \int_0^1 \int_0^{x+y} xy dz dy dx$$

Step 2: Integrate with respect to z

$$\int_0^{x+y} xy dz = xy[z]_0^{x+y} = xy(x+y) = x^2y + xy^2$$

Step 3: Integrate with respect to y

$$\int_0^1 (x^2y + xy^2) dy = \left[x^2 \frac{y^2}{2} + x \frac{y^3}{3} \right]_0^1 = \frac{x^2}{2} + \frac{x}{3}$$

Step 4: Integrate with respect to x

$$I = \int_0^1 \left(\frac{x^2}{2} + \frac{x}{3} \right) dx = \left[\frac{x^3}{6} + \frac{x^2}{6} \right]_0^1 = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}$$

Mark Distribution: • Setting up iterated integral bounds: **1 Mark**

• First integration with respect to z : **1.5 Marks**

• Second integration with respect to y : **1 Mark**

• Final integration yielding $1/3$: **1.5 Marks**

3.3 Continuous Assessment 2 (CA-2) — Set C

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Course Code: MTH-165 **Course Title:** Mathematics for Engineers

Academic Task Type: Continuous Assessment (CA-2) **Set:** C

Maximum Marks: 30 Marks **Time Allowed:** 50 Minutes

Roll No.:

Reg. No.:

Section:

Course Outcomes Addressed: CO4 (Multivariable Limits, Continuity, Partial Derivatives & Optimization), CO5 (Multiple Integrals & Transformations)

Note: Attempt all six questions. Each question carries 5 marks.

Q.1 If $u = \sin^{-1} \left(\frac{x+2y+3z}{\sqrt{x^8+y^8+z^8}} \right)$, verify **Euler's Theorem on Homogeneous Functions** and prove that: **CO4** L3 Apply **5 Marks**

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = -3 \tan u$$

Q.2 Compute the **Jacobian** $\frac{\partial(u,v,w)}{\partial(x,y,z)}$ for the transformation: **CO4** L3 Apply **5 Marks**

$$u = \frac{yz}{x}, \quad v = \frac{zx}{y}, \quad w = \frac{xy}{z}$$

Q.3 Using the **Method of Lagrange Multipliers**, find the shortest distance from the origin $(0, 0, 0)$ to the plane: **CO4** L3 Apply **5 Marks**

$$2x + 3y - z = 14$$

Q.4 Change the order of integration and evaluate: **CO5** L3 Apply **5 Marks**

$$\int_0^1 \int_x^1 \frac{\sin y}{y} dy dx$$

Q.5

Using double integration, compute the total area of the planar region enclosed between the two intersecting parabolas:

CO5

L3 Apply

**5
Marks**

$$y^2 = 4ax \quad \text{and} \quad x^2 = 4ay \quad (a > 0)$$

Q.6

Using **spherical polar coordinates**, evaluate the triple integral:

CO5

L3 Apply

**5
Marks**

$$\iiint_V (x^2 + y^2 + z^2) dV$$

where V is the solid sphere of radius R centered at the origin: $x^2 + y^2 + z^2 \leq R^2$.

Solutions & Marking Scheme (CA-2 Set C)

Step-by-Step Mathematical Solution Solution to Question 1 (5 Marks)

Step 1: Identifying the Homogeneous Function Given $u = \sin^{-1} \left(\frac{x+2y+3z}{\sqrt{x^8+y^8+z^8}} \right)$. Let $f(x, y, z) = \sin u = \frac{x+2y+3z}{\sqrt{x^8+y^8+z^8}}$. Test homogeneity by replacing x, y, z with tx, ty, tz :

$$f(tx, ty, tz) = \frac{t(x+2y+3z)}{\sqrt{t^8(x^8+y^8+z^8)}} = \frac{t(x+2y+3z)}{t^4\sqrt{x^8+y^8+z^8}} = t^{1-4}f(x, y, z) = t^{-3}f(x, y, z)$$

Thus, $f = \sin u$ is a homogeneous function of degree $n = -3$.

Step 2: Applying Euler's Theorem By Euler's Theorem for homogeneous functions:

$$x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} + z \frac{\partial f}{\partial z} = nf = -3f$$

Since $f = \sin u$, by the chain rule:

$$\frac{\partial f}{\partial x} = \cos u \frac{\partial u}{\partial x}, \quad \frac{\partial f}{\partial y} = \cos u \frac{\partial u}{\partial y}, \quad \frac{\partial f}{\partial z} = \cos u \frac{\partial u}{\partial z}$$

Substitute into Euler's equation:

$$x \left(\cos u \frac{\partial u}{\partial x} \right) + y \left(\cos u \frac{\partial u}{\partial y} \right) + z \left(\cos u \frac{\partial u}{\partial z} \right) = -3 \sin u$$

Divide both sides by $\cos u$ (since $\cos u \neq 0$):

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = -3 \frac{\sin u}{\cos u} = -3 \tan u \quad (\text{Hence Proved!})$$

Mark Distribution: • Identifying $f = \sin u$ and testing degree of homogeneity $n = -3$: **2 Marks**

• Stating Euler's Theorem for f : **1.5 Marks**

• Chain rule application and dividing by $\cos u$ to obtain $-3 \tan u$: **1.5 Marks**

Step-by-Step Mathematical Solution Solution to Question 2 (5 Marks)

Step 1: Jacobian Matrix Definition The Jacobian is the determinant of the partial derivative matrix:

$$J = \frac{\partial(u, v, w)}{\partial(x, y, z)} = \begin{vmatrix} u_x & u_y & u_z \\ v_x & v_y & v_z \\ w_x & w_y & w_z \end{vmatrix}$$

Step 2: Partial Derivatives For $u = \frac{yz}{x}$: $u_x = -\frac{yz}{x^2}$, $u_y = \frac{z}{x}$, $u_z = \frac{y}{x}$.

For $v = \frac{zx}{y}$: $v_x = \frac{z}{y}$, $v_y = -\frac{zx}{y^2}$, $v_z = \frac{x}{y}$.

For $w = \frac{xy}{z}$: $w_x = \frac{y}{z}$, $w_y = \frac{x}{z}$, $w_z = -\frac{xy}{z^2}$.

Step 3: Evaluating the Determinant

$$J = \begin{vmatrix} -\frac{yz}{x^2} & \frac{z}{x} & \frac{y}{x} \\ \frac{z}{y} & -\frac{zx}{y^2} & \frac{x}{y} \\ \frac{y}{z} & \frac{x}{z} & -\frac{xy}{z^2} \end{vmatrix}$$

Factor out $\frac{1}{x}$ from row 1, $\frac{1}{y}$ from row 2, and $\frac{1}{z}$ from row 3:

$$J = \frac{1}{xyz} \begin{vmatrix} -\frac{yz}{x} & z & y \\ z & -\frac{zx}{y} & x \\ y & x & -\frac{xy}{z} \end{vmatrix}$$

Now factor out $\frac{1}{x}$ from col 1, $\frac{1}{y}$ from col 2, and $\frac{1}{z}$ from col 3:

$$J = \frac{1}{(xyz)^2} \begin{vmatrix} -yz & yz & yz \\ xz & -xz & xz \\ xy & xy & -xy \end{vmatrix}$$

Factor out yz from row 1, xz from row 2, and xy from row 3:

$$J = \frac{(yz)(xz)(xy)}{(xyz)^2} \begin{vmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix} = \frac{(xyz)^2}{(xyz)^2} \begin{vmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix}$$

Evaluate the 3×3 numerical determinant:

$$\begin{vmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix} = -1((-1)(-1) - 1(1)) - 1(1(-1) - 1(1)) + 1(1(1) - (-1)(1)) \\ = -1(1 - 1) - 1(-1 - 1) + 1(1 + 1) = 0 - 1(-2) + 1(2) = 2 + 2 = 4$$

Therefore:

$$\frac{\partial(\mathbf{u}, \mathbf{v}, \mathbf{w})}{\partial(\mathbf{x}, \mathbf{y}, \mathbf{z})} = 4$$

Mark Distribution: • Partial derivatives setup: **1.5 Marks**

• Determinant formulation and factoring of terms: **2 Marks**

• Numerical determinant evaluation giving final value 4: **1.5 Marks**

Step-by-Step Mathematical Solution Solution to Question 3 (5 Marks)

Step 1: Objective Function and Constraint Distance squared from $(0, 0, 0)$ to (x, y, z) is:

$$f(x, y, z) = x^2 + y^2 + z^2$$

Subject to the constraint $g(x, y, z) = 2x + 3y - z - 14 = 0$.

Step 2: Lagrange Multiplier System Set $\nabla f = \lambda \nabla g$:

$$2x = 2\lambda \implies x = \lambda$$

$$2y = 3\lambda \implies y = \frac{3}{2}\lambda$$

$$2z = -\lambda \implies z = -\frac{1}{2}\lambda$$

Step 3: Finding λ Substitute x, y, z into the constraint:

$$2(\lambda) + 3\left(\frac{3}{2}\lambda\right) - \left(-\frac{1}{2}\lambda\right) = 14$$

$$2\lambda + \frac{9}{2}\lambda + \frac{1}{2}\lambda = 14 \implies 7\lambda = 14 \implies \lambda = 2$$

Thus, the closest point on the plane is:

$$x = 2, \quad y = \frac{3}{2}(2) = 3, \quad z = -\frac{1}{2}(2) = -1 \implies (2, 3, -1)$$

Step 4: Shortest Distance

$$d = \sqrt{x^2 + y^2 + z^2} = \sqrt{2^2 + 3^2 + (-1)^2} = \sqrt{4 + 9 + 1} = \sqrt{14}$$

Mark Distribution: • Objective function and constraint setup: **1 Mark**

• Lagrange equations $\nabla f = \lambda \nabla g$: **1.5 Marks**

• Solving for $\lambda = 2$ and closest point $(2, 3, -1)$: **1.5 Marks**

• Distance computation giving $\sqrt{14}$: **1 Mark**

Step-by-Step Mathematical Solution Solution to Question 4 (5 Marks)

Step 1: Original Region of Integration The integral is:

$$I = \int_0^1 \int_x^1 \frac{\sin y}{y} dy dx$$

Notice that $\int \frac{\sin y}{y} dy$ is non-elementary. Changing order makes it immediately solvable! Given bounds:

$$x \in [0, 1], \quad y \in [x, 1]$$

This represents a triangular region bounded by $y = x$, the line $y = 1$, and the y-axis ($x = 0$).

Step 2: Reversing Order (Horizontal Strips) In the reversed order:

$$y \in [0, 1], \quad x \in [0, y]$$

Thus:

$$I = \int_0^1 \int_0^y \frac{\sin y}{y} dx dy$$

Step 3: Evaluating the Integral Inner integral with respect to x :

$$\int_0^y \frac{\sin y}{y} dx = \frac{\sin y}{y} [x]_0^y = \frac{\sin y}{y} (y) = \sin y$$

Outer integral with respect to y :

$$I = \int_0^1 \sin y dy = [-\cos y]_0^1 = -\cos(1) - (-\cos(0)) = 1 - \cos(1)$$

Mark Distribution: • Region identification and motivation for change of order: **1.5 Marks**

• Correct reversed limits ($y \in [0, 1], x \in [0, y]$): **1.5 Marks**

• Evaluation of inner and outer integrals giving $1 - \cos(1)$: **2 Marks**

Step-by-Step Mathematical Solution Solution to Question 5 (5 Marks)

Step 1: Intersection of the Parabolas Given $y^2 = 4ax$ and $x^2 = 4ay \implies y = \frac{x^2}{4a}$. Substitute y into the first equation:

$$\left(\frac{x^2}{4a}\right)^2 = 4ax \implies \frac{x^4}{16a^2} = 4ax \implies x(x^3 - 64a^3) = 0$$

The points of intersection are $x = 0$ and $x = 4a$. At $x = 0 \implies y = 0$. At $x = 4a \implies y = 4a$. Vertices: $(0, 0)$ and $(4a, 4a)$.

Step 2: Area Formulation via Double Integration The area of the enclosed domain is:

$$A = \iint_R dx dy = \int_0^{4a} \int_{x^2/(4a)}^{2\sqrt{ax}} dy dx$$

Step 3: Evaluation

$$\begin{aligned} A &= \int_0^{4a} \left(2\sqrt{ax}^{1/2} - \frac{x^2}{4a} \right) dx = \left[2\sqrt{a} \frac{x^{3/2}}{3/2} - \frac{x^3}{12a} \right]_0^{4a} \\ &= \frac{4\sqrt{a}}{3} (4a)^{3/2} - \frac{(4a)^3}{12a} = \frac{4\sqrt{a}}{3} (8a^{3/2}) - \frac{64a^3}{12a} = \frac{32a^2}{3} - \frac{16a^2}{3} = \frac{16a^2}{3} \end{aligned}$$

Mark Distribution: • Intersection points calculation: **1.5 Marks**

• Double integral setup with correct limits: **1.5 Marks**

• Accurate integration giving $16a^2/3$: **2 Marks**

Step-by-Step Mathematical Solution Solution to Question 6 (5 Marks)**Step 1: Spherical Polar Coordinates** Let:

$$x = \rho \sin \phi \cos \theta, \quad y = \rho \sin \phi \sin \theta, \quad z = \rho \cos \phi$$

Then $x^2 + y^2 + z^2 = \rho^2$, and the volume element is:

$$dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$$

Step 2: Domain Limits for Full Sphere of Radius R

$$\rho \in [0, R], \quad \phi \in [0, \pi], \quad \theta \in [0, 2\pi]$$

Step 3: Integral Evaluation The three variables are completely decoupled:

$$\iiint_V (x^2 + y^2 + z^2) \, dV = \int_0^{2\pi} d\theta \cdot \int_0^\pi \sin \phi \, d\phi \cdot \int_0^R \rho^2 (\rho^2 \, d\rho)$$

Evaluate each factor:

- $\int_0^{2\pi} d\theta = 2\pi$
- $\int_0^\pi \sin \phi \, d\phi = [-\cos \phi]_0^\pi = -(-1) - (-1) = 2$
- $\int_0^R \rho^4 \, d\rho = \left[\frac{\rho^5}{5}\right]_0^R = \frac{R^5}{5}$

Multiplying the three components:

$$I = (2\pi)(2) \left(\frac{R^5}{5}\right) = \frac{4\pi R^5}{5}$$

Mark Distribution: • Spherical coordinate equations and Jacobian $\rho^2 \sin \phi$: **1.5 Marks**• Correct limits for complete sphere: **1.5 Marks**• Evaluation of each single integral giving $4\pi R^5/5$: **2 Marks**

Chapter 4

End-Term Examination Suite (70 Marks Comprehensive)

The End-Term Examination is the comprehensive university final examination (Total = **70 Marks**, Time Allowed = **3 Hours**):

- **Part A (Objective — 30 Marks):** 30 MCQs strictly testing **Unit 4, Unit 5, and Unit 6** (10 MCQs from each unit, 1 mark each, negative marking -0.25).
- **Part B (Descriptive — 40 Marks):** 5 comprehensive questions of 10 marks each (3 questions from Units 4, 5, 6 and 2 questions from Units 1–6). Students must **attempt any 4 questions** ($4 \times 10 = 40$ Marks).

4.1 End-Term Examination — Set A

LOVELY PROFESSIONAL UNIVERSITY

Transforming Education, Transforming India

Course Code: MTH-165 **Course Title:** Mathematics for Engineers

Academic Task Type: End-Term Examination **Set:** A

Maximum Marks: 60 Marks **Time Allowed:** 3 Hours

Course Outcomes Addressed: CO1 to CO6 (Units 1 to 6 Comprehensive)

Roll No.:

Reg. No.:

Section:

Structure of Question Paper:

Part A (Objective): 30 MCQs strictly from **Units 4, 5, and 6**. Total weight: **20 Marks**. All questions compulsory.

Part B (Subjective): 6 Questions from **all units** (10 Marks each). **Attempt any 4 questions** ($4 \times 10 = 40$ Marks).

Total Maximum Marks: 60 Marks.

PART A: Objective Section (30 MCQs — Units 4, 5, 6) [20 Marks]

Q.1 The value of $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{\sqrt{x^2+y^2}}$ is:

[1 Mark]

- (A) 0 (C) Does not exist
(B) 1 (D) ∞

Q.2 If $u = \ln(x^3 + y^3 + z^3 - 3xyz)$, then $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z}$ is:

[1 Mark]

- (A) $\frac{3}{x+y+z}$ (C) $\frac{9}{x+y+z}$
(B) $\frac{1}{x+y+z}$ (D) 0

Q.3 If $f(x, y)$ is homogeneous of degree n , then $x^2 f_{xx} + 2xy f_{xy} + y^2 f_{yy}$ equals:

[1 Mark]

- (A) $n(n-1)f$ (C) nf
(B) $n^2 f$ (D) $(n+1)f$

Q.4 The Jacobian $\frac{\partial(x,y)}{\partial(r,\theta)}$ for polar coordinates $x = r \cos \theta$, $y = r \sin \theta$ is:

[1 Mark]

- (A) r (C) $1/r$
(B) r^2 (D) 1

Q.5 At a critical point (a, b) , if $rt - s^2 < 0$, the point is a:

[1 Mark]

- (A) Local Minimum (C) Saddle Point
(B) Local Maximum (D) Point of Inflexion

Q.6 The directional derivative of $\phi = xy + yz + zx$ at $(1, 2, 0)$ in the direction of $\hat{i} + 2\hat{j} + 2\hat{k}$ is: [1 Mark]

- (A) $\frac{10}{3}$ (C) $\frac{8}{3}$
 (B) 3 (D) 4

Q.7 A vector field $\vec{F}$ is called **solenoidal** if: [1 Mark]

- (A) $\nabla \times \vec{F} = 0$ (C) $\nabla^2 \vec{F} = 0$
 (B) $\nabla \cdot \vec{F} = 0$ (D) $\vec{F} \cdot \vec{r} = 0$

Q.8 A vector field $\vec{F}$ is called **irrotational** if: [1 Mark]

- (A) $\nabla \cdot \vec{F} = 0$ (C) $\nabla \phi = \vec{F}$
 (B) $\nabla \times \vec{F} = \vec{0}$ (D) $\nabla^2 \vec{F} = 0$

Q.9 The value of $\nabla \cdot \vec{r}$ where $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ is: [1 Mark]

- (A) 0 (C) 2
 (B) 1 (D) 3

Q.10 The value of $\nabla \times \vec{r}$ is: [1 Mark]

- (A) $\vec{0}$ (C) $\vec{r}$
 (B) 3 (D) 1

Q.11 The value of the double integral $\int_0^1 \int_0^2 xy \, dx \, dy$ is: [1 Mark]

- (A) 1 (C) $1/2$
 (B) 2 (D) 4

Q.12 The area of a region R in the xy -plane is represented by: [1 Mark]

- (A) $\iint_R dx \, dy$ (C) $\iint_R (x + y) \, dx \, dy$
 (B) $\iint_R xy \, dx \, dy$ (D) $\oint_C x \, dx$

Q.13 In spherical polar coordinates, the volume element dV is: [1 Mark]

- (A) $\rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$ (C) $\rho^2 \cos \phi \, d\rho \, d\phi \, d\theta$
 (B) $\rho \sin \phi \, d\rho \, d\phi \, d\theta$ (D) $\rho^2 \, d\rho \, d\phi \, d\theta$

Q.14 Green's Theorem in a plane converts a line integral around a closed curve into a: [1 Mark]

- (A) Surface Integral (C) Triple Integral
 (B) Double Integral over the enclosed region (D) Volume Integral

Q.15 Stokes' Theorem relates a line integral around a boundary curve C to a: [1 Mark]

- (A) Volume Integral (C) Double Integral of $\nabla \cdot \vec{F}$
 (B) Surface Integral of $\nabla \times \vec{F}$ (D) Triple Integral

Q.16 Gauss Divergence Theorem relates a surface integral over a closed surface S to a: [1 Mark]

- (A) Line Integral (C) Surface Integral of Curl
 (B) Volume Integral of $\nabla \cdot \vec{F}$ (D) Double Integral

Q.17 The value of $\nabla(r^n)$ where $r = \|\vec{r}\|$ is: [1 Mark]

- (A) $nr^{n-1}\vec{r}$ (C) $r^n\vec{r}$
 (B) $nr^{n-2}\vec{r}$ (D) nr^n

Q.18 The value of $\nabla^2(1/r)$ for $r \neq 0$ is: [1 Mark]

- (A) 0 (C) $-1/r^3$
 (B) $1/r^2$ (D) 4π

Q.19 If $\vec{F} = \nabla\phi$, then the work done in moving a particle from point A to point B in the field is: [1 Mark]

- (A) $\phi(B) - \phi(A)$ (C) $\phi(A)\phi(B)$
 (B) $\phi(A) - \phi(B)$ (D) 0

Q.20 The value of the integral $\int_0^{\pi/2} \int_0^{\pi/2} \sin(x+y) dx dy$ is: [1 Mark]

- (A) 1 (C) 0
 (B) 2 (D) π

Q.21 Changing the order of integration of $\int_0^1 \int_0^x f(x,y) dy dx$ gives: [1 Mark]

- (A) $\int_0^1 \int_y^1 f(x,y) dx dy$ (C) $\int_0^x \int_0^1 f(x,y) dx dy$
 (B) $\int_0^1 \int_0^y f(x,y) dx dy$ (D) $\int_y^1 \int_0^1 f(x,y) dx dy$

Q.22 The total volume of a sphere of radius a computed via triple integration is: [1 Mark]

- (A) $\frac{4}{3}\pi a^3$ (C) $\frac{2}{3}\pi a^3$
 (B) $4\pi a^2$ (D) πa^3

Q.23 The curl of the gradient of any scalar field ϕ is identically: [1 Mark]

- (A) $\vec{0}$ (C) $\nabla^2\phi$
 (B) 1 (D) $\vec{r}$

Q.24 The divergence of the curl of any vector field $\vec{F}$ is identically: [1 Mark]

- (A) 0 (C) $\nabla^2 \vec{F}$
(B) 1 (D) $\vec{F}$

Q.25 A function $f(x, y)$ has a local minimum at (a, b) if: [1 Mark]

- (A) $rt - s^2 > 0$ and $r > 0$ (C) $rt - s^2 < 0$
(B) $rt - s^2 > 0$ and $r < 0$ (D) $rt - s^2 = 0$

Q.26 The unit normal vector to the surface $\phi(x, y, z) = c$ at a point is given by: [1 Mark]

- (A) $\frac{\nabla \phi}{\|\nabla \phi\|}$ (C) $\frac{\vec{r}}{\|\vec{r}\|}$
(B) $\nabla \phi$ (D) $\nabla \times \vec{r}$

Q.27 The value of $\oint_C (x dy - y dx)$ over a closed curve enclosing area A is: [1 Mark]

- (A) A (C) 0
(B) $2A$ (D) $-A$

Q.28 If $u = x + y$ and $v = x - y$, the Jacobian $\frac{\partial(u,v)}{\partial(x,y)}$ is: [1 Mark]

- (A) 2 (C) 0
(B) -2 (D) 1

Q.29 The integral $\iint_R r dr d\theta$ represents the: [1 Mark]

- (A) Volume (C) Circumference
(B) Area of region R (D) Moment of Inertia

Q.30 The value of $\iint_S \vec{r} \cdot \hat{n} dS$ over a closed surface bounding volume V is: [1 Mark]

- (A) V (C) $3V$
(B) $2V$ (D) 0
-

PART B: Descriptive Section (Attempt Any 4 of 6) [40 Marks]

- Q.31** Find the eigenvalues and a complete set of linearly independent eigenvectors of the symmetric matrix: **CO1** L3 Apply **10 Marks**
(Unit 1)

$$A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$$

Verify that the eigenvectors corresponding to distinct eigenvalues are mutually orthogonal.

- Q.32** Solve the third-order non-homogeneous ordinary differential equation by the operator method: **CO2** L3 Apply **10 Marks**
(Unit 2)

$$(D^3 - 3D^2 + 4)y = e^{2x} + 40 \cos x$$

- Q.33** (a) Test the convergence of the infinite positive series: **CO3** L4 Analyze **10 Marks**
(Unit 3)

$$\sum_{n=1}^{\infty} \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots (2n)} \cdot \frac{1}{2n+1}$$

(b) Determine whether the alternating series $\sum_{n=2}^{\infty} \frac{(-1)^n}{n(\ln n)^2}$ converges absolutely or conditionally.

- Q.34** A rectangular box open at the top is to have a volume of 32 cm^3 . Find the dimensions of the box that require the **minimum surface area** of material, using the **Method of Lagrange Multipliers**. **CO4** L3 Apply **10 Marks**
(Unit 4)

- Q.35** (a) Change the order of integration and evaluate: **CO5** L3 Apply **10 Marks**
(Unit 5)

$$\int_0^1 \int_{\sqrt{y}}^1 \sqrt{x^3 + 1} \, dx \, dy$$

(b) Evaluate $\iiint_V z \, dV$ where V is the solid upper hemisphere $x^2 + y^2 + z^2 \leq a^2$ ($z \geq 0$).

- Q.36** Verify the **Gauss Divergence Theorem** for the vector field: **CO6** L4 Analyze **10 Marks**
(Unit 6)

$$\vec{F} = 4x\hat{i} - 2y^2\hat{j} + z^2\hat{k}$$

over the cylindrical solid bounded by $x^2 + y^2 = 4$, $z = 0$, and $z = 3$.

Solutions & Marking Scheme (End-Term Set A)

Part A: Answer Key & Explanations

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	6	A	11	A	16	B	21	A	26	A
2	A	7	B	12	A	17	B	22	A	27	B
3	A	8	B	13	A	18	A	23	A	28	B
4	A	9	D	14	B	19	A	24	A	29	B
5	C	10	A	15	B	20	B	25	A	30	C

Step-by-Step Mathematical Solution Explanations for Part A

Q.1 (Ans: A) Let $x = r \cos \theta, y = r \sin \theta$: $\frac{xy}{\sqrt{x^2+y^2}} = \frac{r^2 \cos \theta \sin \theta}{r} = r \cos \theta \sin \theta \rightarrow 0$ as $r \rightarrow 0$.

Q.4 (Ans: A) $J = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r(\cos^2 \theta + \sin^2 \theta) = r$.

Q.9 (Ans: D) $\nabla \cdot \vec{r} = \frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial z}{\partial z} = 1 + 1 + 1 = 3$.

Q.10 (Ans: A) $\nabla \times \vec{r} = \vec{0}$ because the position vector field is purely irrotational.

Q.17 (Ans: B) $\nabla(r^n) = nr^{n-1}\nabla r = nr^{n-1}\frac{\vec{r}}{r} = nr^{n-2}\vec{r}$.

Q.30 (Ans: C) By Gauss Divergence Theorem, $\iint_S \vec{r} \cdot \hat{n} dS = \iiint_V (\nabla \cdot \vec{r}) dV = \iiint_V 3 dV = 3V$.

Part B: Detailed Subjective Solutions

Step-by-Step Mathematical Solution Solution to Question 31 (10 Marks)

Step 1: Characteristic Equation $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$.

- $S_1 = \text{trace}(A) = 6 + 3 + 3 = 12$.
- $S_2 = M_{11} + M_{22} + M_{33} = (9 - 1) + (18 - 4) + (18 - 4) = 8 + 14 + 14 = 36$.
- $S_3 = \det(A) = 6(8) - (-2)(-6 + 2) + 2(2 - 6) = 48 + 2(-4) + 2(-4) = 48 - 8 - 8 = 32$.

Equation: $\lambda^3 - 12\lambda^2 + 36\lambda - 32 = 0$. Factorization: $(\lambda - 2)(\lambda - 2)(\lambda - 8) = 0 \implies \lambda_1 = 8, \lambda_2 = 2, \lambda_3 = 2$.

Step 2: Eigenvector for $\lambda = 8$ $(A - 8I)X = 0 \implies \begin{bmatrix} -2 & -2 & 2 \\ -2 & -5 & -1 \\ 2 & -1 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$. From

row 1: $-x_1 - x_2 + x_3 = 0$. Using cross-multiplication on first two rows: $\frac{x_1}{(-2)(-1) - (-2)(-5)} = \frac{x_2}{(2)(-2) - (-2)(-1)} = \frac{x_3}{(-2)(-5) - (-2)(-1)}$

$\frac{x_2}{(2)(-2) - (-2)(-1)} = \frac{x_3}{(-2)(-5) - (-2)(-1)} \implies \frac{x_1}{12} = \frac{x_2}{-6} = \frac{x_3}{6}$. Dividing by 6 gives: $\mathbf{X}_1 = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$.

Step 3: Eigenvectors for Repeated Root $\lambda = 2$ $(A - 2I)X = 0 \implies \begin{bmatrix} 4 & -2 & 2 \\ -2 & 1 & -1 \\ 2 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$.

All three rows are scalar multiples of $2x_1 - x_2 + x_3 = 0 \implies x_2 = 2x_1 + x_3$. Here we have two free variables (x_1 and x_3):

- Let $x_1 = 1, x_3 = 0 \implies x_2 = 2$: $\mathbf{X}_2 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$.
- Let $x_1 = 0, x_3 = 1 \implies x_2 = 1$: $\mathbf{X}_3' = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$.

To ensure X_3 is orthogonal to X_2 (Gram-Schmidt): Let $X_3 = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$ satisfy $2a - b + c = 0$ and $X_2 \cdot X_3 = a + 2b = 0 \implies a = -2b$. Then $2(-2b) - b + c = 0 \implies c = 5b$. Choosing $b = 1$:

$\mathbf{X}_3 = \begin{bmatrix} -2 \\ 1 \\ 5 \end{bmatrix}$.

Step 4: Orthogonality Verification $X_1 \cdot X_2 = (2)(1) + (-1)(2) + (1)(0) = 2 - 2 + 0 = 0$. $X_1 \cdot X_3 = (2)(-2) + (-1)(1) + (1)(5) = -4 - 1 + 5 = 0$. $X_2 \cdot X_3 = (1)(-2) + (2)(1) + (0)(5) = -2 + 2 + 0 = 0$. All three eigenvectors are mutually orthogonal.

Mark Distribution: • Characteristic equation and roots $\lambda = 8, 2, 2$: **3 Marks**

- Eigenvector for $\lambda = 8$: **2 Marks**
- Linearly independent eigenvectors for $\lambda = 2$: **3 Marks**
- Proof of mutual orthogonality: **2 Marks**

Step-by-Step Mathematical Solution Solution to Question 32 (10 Marks)

Step 1: Complementary Function Auxiliary equation: $m^3 - 3m^2 + 4 = 0$. Test $m = -1$:
 $(-1)^3 - 3(-1)^2 + 4 = -1 - 3 + 4 = 0 \implies (m + 1)$ is a factor. $(m + 1)(m^2 - 4m + 4) = 0 \implies$
 $(m + 1)(m - 2)^2 = 0 \implies m = -1, 2, 2$.

$$y_c = c_1 e^{-x} + (c_2 + c_3 x) e^{2x}$$

Step 2: Particular Integral for e^{2x} (y_{p1}) Since $m = 2$ is a root of multiplicity 2:

$$y_{p1} = \frac{1}{(D+1)(D-2)^2} e^{2x} = \frac{1}{(2+1)} \frac{1}{(D-2)^2} e^{2x} = \frac{1}{3} \frac{x^2}{2!} e^{2x} = \frac{1}{6} x^2 e^{2x}$$

Step 3: Particular Integral for $40 \cos x$ (y_{p2})

$$y_{p2} = \frac{1}{D^3 - 3D^2 + 4} (40 \cos x) = \frac{40}{D(D^2 - 3D^2 + 4)} \cos x$$

Replace D^2 with $-1^2 = -1$:

$$= \frac{40}{D(-1) - 3(-1) + 4} \cos x = \frac{40}{-D + 7} \cos x = \frac{40}{7 - D} \cos x$$

Multiply numerator and denominator by $(7 + D)$:

$$= \frac{40(7 + D)}{49 - D^2} \cos x = \frac{40(7 + D)}{49 - (-1)} \cos x = \frac{40(7 + D)}{50} \cos x = \frac{4}{5} (7 \cos x - \sin x)$$

Step 4: General Solution

$$y(x) = c_1 e^{-x} + (c_2 + c_3 x) e^{2x} + \frac{1}{6} x^2 e^{2x} + \frac{28}{5} \cos x - \frac{4}{5} \sin x$$

Mark Distribution: • Auxiliary equation and roots $m = -1, 2, 2$: **3 Marks**

• Particular integral y_{p1} with repeated root handling: **3 Marks**

• Particular integral y_{p2} for cosine: **3 Marks**

• Complete general solution: **1 Mark**

Step-by-Step Mathematical Solution Solution to Question 33 (10 Marks)

Part (a): Convergence of the Positive Term Series (5 Marks) Let $u_n = \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots (2n)} \cdot \frac{1}{2n+1}$. Then:

$$u_{n+1} = \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)(2n+1)}{2 \cdot 4 \cdot 6 \cdots (2n)(2n+2)} \cdot \frac{1}{2n+3}$$

Apply D'Alembert's Ratio Test:

$$\frac{u_n}{u_{n+1}} = \frac{2n+2}{2n+1} \cdot \frac{2n+3}{2n+1} = \frac{4n^2 + 10n + 6}{4n^2 + 4n + 1}$$

$\lim_{n \rightarrow \infty} \frac{u_n}{u_{n+1}} = 1$. The Ratio Test is **inconclusive!** Apply **Raabe's Test**:

$$\begin{aligned} L &= \lim_{n \rightarrow \infty} n \left(\frac{u_n}{u_{n+1}} - 1 \right) = \lim_{n \rightarrow \infty} n \left(\frac{4n^2 + 10n + 6 - 4n^2 - 4n - 1}{4n^2 + 4n + 1} \right) \\ &= \lim_{n \rightarrow \infty} n \left(\frac{6n + 5}{4n^2 + 4n + 1} \right) = \lim_{n \rightarrow \infty} \frac{6n^2 + 5n}{4n^2 + 4n + 1} = \frac{6}{4} = \frac{3}{2} \end{aligned}$$

Since $L = \frac{3}{2} > 1$, by Raabe's Test, the series is **strictly CONVERGENT**.

Part (b): Alternating Series Convergence (5 Marks) Given $\sum_{n=2}^{\infty} (-1)^n a_n$ where $a_n = \frac{1}{n(\ln n)^2}$.

- (i) Consider the series of absolute terms: $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$. Apply Cauchy's Integral Test with $f(x) = \frac{1}{x(\ln x)^2}$:

$$\int_2^{\infty} \frac{dx}{x(\ln x)^2} = \left[-\frac{1}{\ln x} \right]_2^{\infty} = 0 - \left(-\frac{1}{\ln 2} \right) = \frac{1}{\ln 2} < \infty$$

The integral converges $\implies$ The series of absolute values converges!

- (ii) Since the absolute series converges, the given alternating series is **ABSOLUTELY CONVERGENT**.

Mark Distribution: • Part (a) Ratio test failure ($L = 1$): **2 Marks**

• Part (a) Raabe's test evaluation showing $L = 3/2 > 1$ (Convergent): **3 Marks**

• Part (b) Integral test on absolute series: **3 Marks**

• Part (b) Conclusion of absolute convergence: **2 Marks**

Step-by-Step Mathematical Solution Solution to Question 34 (10 Marks)

Step 1: Mathematical Formulation Let x, y be the horizontal dimensions of the base and z be the height. The box is open at the top, so its surface area consists of the base and 4 sides:

$$S(x, y, z) = xy + 2xz + 2yz$$

Subject to the volume constraint:

$$g(x, y, z) = xyz - 32 = 0$$

Step 2: Lagrange Multiplier Equations Set $\nabla S = \lambda \nabla g$:

$$S_x = \lambda g_x \implies y + 2z = \lambda yz \quad (4.1)$$

$$S_y = \lambda g_y \implies x + 2z = \lambda xz \quad (4.2)$$

$$S_z = \lambda g_z \implies 2x + 2y = \lambda xy \quad (4.3)$$

Step 3: Solving the System Multiply (4.1) by x and (4.2) by y :

$$xy + 2xz = \lambda xyz$$

$$xy + 2yz = \lambda xyz$$

Equating both gives:

$$xy + 2xz = xy + 2yz \implies 2xz = 2yz \implies \mathbf{x = y} \quad (\text{since } z \neq 0)$$

Now substitute $x = y$ into (4.3):

$$2x + 2x = \lambda x^2 \implies 4x = \lambda x^2 \implies \lambda = \frac{4}{x}$$

Substitute $\lambda = \frac{4}{x}$ into (4.1) with $y = x$:

$$x + 2z = \left(\frac{4}{x}\right)xz = 4z \implies x = 2z \implies \mathbf{z = \frac{x}{2}}$$

Step 4: Using Volume Constraint Substitute $y = x$ and $z = x/2$ into $xyz = 32$:

$$(x)(x)\left(\frac{x}{2}\right) = 32 \implies \frac{x^3}{2} = 32 \implies x^3 = 64 \implies \mathbf{x = 4 \text{ cm}}$$

Hence:

$$\mathbf{y = 4 \text{ cm}, \quad z = 2 \text{ cm}}$$

Step 5: Minimum Surface Area

$$S_{\min} = (4)(4) + 2(4)(2) + 2(4)(2) = 16 + 16 + 16 = \mathbf{48 \text{ cm}^2}$$

Mark Distribution: • Objective function S and constraint g : **2 Marks**

• Lagrange system setup $\nabla S = \lambda \nabla g$: **3 Marks**

• Solving relation $x = y = 2z$: **3 Marks**

• Final dimensions $4 \times 4 \times 2$ cm and $S = 48 \text{ cm}^2$: **2 Marks**

Step-by-Step Mathematical Solution Solution to Question 35 (10 Marks)

Part (a): Change of Order of Integration (5 Marks) Given: $I = \int_0^1 \int_{\sqrt{y}}^1 \sqrt{x^3 + 1} dx dy$.
Limits: $y \in [0, 1]$, $x \in [\sqrt{y}, 1] \implies y = x^2$. In reversed order (vertical strips): $x \in [0, 1]$, $y \in [0, x^2]$.

$$I = \int_0^1 \int_0^{x^2} \sqrt{x^3 + 1} dy dx = \int_0^1 \sqrt{x^3 + 1} [y]_0^{x^2} dx = \int_0^1 x^2 \sqrt{x^3 + 1} dx$$

Let $u = x^3 + 1 \implies du = 3x^2 dx \implies x^2 dx = \frac{du}{3}$. When $x = 0$, $u = 1$; when $x = 1$, $u = 2$:

$$I = \frac{1}{3} \int_1^2 u^{1/2} du = \frac{1}{3} \left[\frac{u^{3/2}}{3/2} \right]_1^2 = \frac{2}{9} (2^{3/2} - 1) = \frac{2}{9} (2\sqrt{2} - 1)$$

Part (b): Triple Integral in Spherical Coordinates (5 Marks) Region V : Upper hemisphere of radius a ($z \geq 0$). In spherical polar coordinates: $z = \rho \cos \phi$, $dV = \rho^2 \sin \phi d\rho d\phi d\theta$. Limits: $\rho \in [0, a]$, $\phi \in [0, \pi/2]$ (upper half), $\theta \in [0, 2\pi]$.

$$\iiint_V z dV = \int_0^{2\pi} d\theta \cdot \int_0^{\pi/2} \cos \phi \sin \phi d\phi \cdot \int_0^a \rho^3 d\rho$$

- $\int_0^{2\pi} d\theta = 2\pi$
- $\int_0^{\pi/2} \sin \phi \cos \phi d\phi = \left[\frac{\sin^2 \phi}{2} \right]_0^{\pi/2} = \frac{1}{2}$
- $\int_0^a \rho^3 d\rho = \frac{a^4}{4}$

Multiplying:

$$\iiint_V z dV = 2\pi \left(\frac{1}{2} \right) \left(\frac{a^4}{4} \right) = \frac{\pi a^4}{4}$$

Mark Distribution: • Part (a) reversed limits and integration yielding $\frac{2}{9}(2\sqrt{2} - 1)$: **5 Marks**

• Part (b) spherical bounds and integration yielding $\frac{\pi a^4}{4}$: **5 Marks**

Step-by-Step Mathematical Solution Solution to Question 36 (10 Marks)**Step 1: Statement of Gauss Divergence Theorem**

$$\iint_S \vec{F} \cdot \hat{n} dS = \iiint_V (\nabla \cdot \vec{F}) dV$$

Given $\vec{F} = 4x\hat{i} - 2y^2\hat{j} + z^2\hat{k}$ over cylinder $x^2 + y^2 \leq 4, 0 \leq z \leq 3$.

Step 2: Volume Integral Evaluation Compute $\nabla \cdot \vec{F}$:

$$\nabla \cdot \vec{F} = \frac{\partial}{\partial x}(4x) + \frac{\partial}{\partial y}(-2y^2) + \frac{\partial}{\partial z}(z^2) = 4 - 4y + 2z$$

Transform to cylindrical coordinates: $x = r \cos \theta, y = r \sin \theta, z = z, dV = r dz dr d\theta$. Bounds: $r \in [0, 2], \theta \in [0, 2\pi], z \in [0, 3]$.

$$\iiint_V (4 - 4r \sin \theta + 2z)r dz dr d\theta$$

Due to periodicity, $\int_0^{2\pi} \sin \theta d\theta = 0$, so the middle term vanishes!

$$\begin{aligned} &= \int_0^{2\pi} d\theta \int_0^2 r dr \int_0^3 (4 + 2z) dz = (2\pi) \left[\frac{r^2}{2} \right]_0^2 \left[4z + z^2 \right]_0^3 \\ &= (2\pi)(2)(12 + 9) = 4\pi(21) = 84\pi \end{aligned}$$

Step 3: Surface Integral Evaluation (3 Surfaces)

(i) **Bottom Surface** S_1 ($z = 0$): $\hat{n} = -\hat{k}, z = 0 \implies \vec{F} \cdot \hat{n} = -z^2 = 0 \implies \iint_{S_1} = 0$.

(ii) **Top Surface** S_2 ($z = 3$): $\hat{n} = \hat{k}, z = 3 \implies \vec{F} \cdot \hat{n} = z^2 = 9$.

$$\iint_{S_2} 9 dS = 9 \times \text{Area}(\text{circle of radius 2}) = 9(\pi \cdot 2^2) = 36\pi$$

(iii) **Curved Surface** S_3 ($r = 2$): Normal $\hat{n} = \frac{x\hat{i} + y\hat{j}}{2}$.

$$\vec{F} \cdot \hat{n} = \frac{4x^2 - 2y^3}{2} = 2x^2 - y^3 = 2(2 \cos \theta)^2 - (2 \sin \theta)^3 = 8 \cos^2 \theta - 8 \sin^3 \theta$$

$dS = 2 d\theta dz$:

$$\iint_{S_3} \vec{F} \cdot \hat{n} dS = \int_0^3 dz \int_0^{2\pi} (8 \cos^2 \theta - 8 \sin^3 \theta) 2 d\theta$$

Since $\int_0^{2\pi} \sin^3 \theta d\theta = 0$ and $\int_0^{2\pi} \cos^2 \theta d\theta = \pi$:

$$= (3)(2)(8\pi) = 48\pi$$

Total surface integral:

$$\iint_S \vec{F} \cdot \hat{n} dS = 0 + 36\pi + 48\pi = 84\pi$$

Since Volume Integral (84π) = Surface Integral (84π), **Gauss Divergence Theorem is verified!**

Mark Distribution: • Stating theorem and computing $\nabla \cdot \vec{F} = 4 - 4y + 2z$: **2 Marks**

• Evaluating volume integral to obtain 84π : **3 Marks**

• Evaluating surface integrals across top, bottom, and curved faces: **4 Marks**

• Equating and concluding verification: **1 Mark**

4.2 End-Term Examination — Set B

LOVELY PROFESSIONAL UNIVERSITY

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Course Code: MTH-165 **Course Title:** Mathematics for Engineers

Academic Task Type: End-Term Examination **Set:** B

Maximum Marks: 60 Marks **Time Allowed:** 3 Hours

Course Outcomes Addressed: CO1 to CO6 (Units 1 to 6 Comprehensive)

Roll No.:

Reg. No.:

Section:

Structure of Question Paper:

Part A (Objective): 30 MCQs strictly from **Units 4, 5, and 6**. Total weight: **20 Marks**. All questions compulsory.

Part B (Subjective): 6 Questions from **all units** (10 Marks each). **Attempt any 4 questions** ($4 \times 10 = 40$ Marks).

Total Maximum Marks: 60 Marks.

PART A: Objective Section (30 MCQs — Units 4, 5, 6) [20 Marks]

Q.1 The function $f(x, y) = \frac{x^2 - y^2}{x^2 + y^2}$ at $(0, 0)$ has: [1 Mark]

- (A) Limit equal to 0 (C) No limit (limit does not exist)
(B) Limit equal to 1 (D) Limit equal to -1

Q.2 If $z = e^{x/y}$, then $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y}$ is: [1 Mark]

- (A) 0 (C) $2z$
(B) z (D) $-z$

Q.3 The value of the Jacobian $\frac{\partial(u, v)}{\partial(x, y)}$ when $u = x^2 - y^2$ and $v = 2xy$ is: [1 Mark]

- (A) $4(x^2 + y^2)$ (C) $x^2 + y^2$
(B) $2(x^2 + y^2)$ (D) $4xy$

Q.4 The maximum value of the directional derivative of ϕ at any point is along the: [1 Mark]

- (A) Tangent vector (C) Normal to the gradient
(B) Gradient vector $\nabla \phi$ (D) Coordinate axis

Q.5 If $\vec{F} = (x + 3y)\hat{i} + (y - 2z)\hat{j} + (x + az)\hat{k}$ is solenoidal, then the constant a is: [1 Mark]

- (A) -2 (C) 2
(B) -1 (D) 0

Q.6 The scalar potential ϕ for the conservative force $\vec{F} = 2xy\hat{i} + x^2\hat{j}$ is: [1 Mark]

- (A) $x^2y + C$ (C) $x^2 + y^2 + C$
(B) $2xy^2 + C$ (D) $xy + C$

Q.7 The value of $\iint_R dx dy$ where R is the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} \leq 1$ is: [1 Mark]

- (A) πab (C) $\frac{\pi ab}{4}$
(B) $4\pi ab$ (D) $2\pi ab$

Q.8 The value of $\int_0^a \int_0^{\sqrt{a^2-x^2}} dy dx$ is: [1 Mark]

- (A) $\frac{\pi a^2}{4}$ (C) $\frac{\pi a^2}{2}$
(B) πa^2 (D) a^2

Q.9 If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ and $r = \|\vec{r}\|$, then $\nabla(1/r)$ is: [1 Mark]

- (A) $-\frac{\vec{r}}{r^3}$ (C) $-\frac{\vec{r}}{r^2}$
(B) $\frac{\vec{r}}{r^3}$ (D) $\frac{1}{r^2}$

Q.10 A vector field $\vec{F}$ is conservative if and only if: [1 Mark]

- (A) $\nabla \times \vec{F} = \vec{0}$ (C) $\nabla^2 \vec{F} = 0$
(B) $\nabla \cdot \vec{F} = 0$ (D) $\vec{F} = 0$

Q.11 The value of $\int_0^1 \int_0^1 \int_0^1 (x + y + z) dx dy dz$ is: [1 Mark]

- (A) $3/2$ (C) 3
(B) 1 (D) $1/2$

Q.12 The stationary points of $f(x, y) = x^2 + y^2$ are: [1 Mark]

- (A) $(0, 0)$ only (C) $(0, 1)$ and $(1, 0)$
(B) $(1, 1)$ only (D) Infinitely many

Q.13 For $f(x, y) = xy$, the origin $(0, 0)$ is a: [1 Mark]

- (A) Local Minimum (C) Saddle Point
(B) Local Maximum (D) Inconclusive

Q.14 The value of $\nabla \times (\nabla \phi)$ for any scalar field ϕ is: [1 Mark]

- (A) $\vec{0}$ (C) $\nabla^2 \phi$
(B) 1 (D) $\vec{r}$

Q.15 The value of the line integral $\oint_C (x dy - y dx)$ along the circle $x^2 + y^2 = 1$ in counter-clockwise direction is: [1 Mark]

- (A) 2π (C) 0
(B) π (D) 4π

Q.16 Stokes' Theorem equates $\oint_C \vec{F} \cdot d\vec{r}$ to: [1 Mark]

- (A) $\iint_S (\nabla \times \vec{F}) \cdot \hat{n} dS$ (C) $\iint_S \vec{F} \cdot \hat{n} dS$
(B) $\iiint_V (\nabla \cdot \vec{F}) dV$ (D) $\oint_C (\nabla \cdot \vec{F}) ds$

Q.17 The value of $\nabla \cdot (\nabla \times \vec{A})$ for any vector field $\vec{A}$ is: [1 Mark]

- (A) 0 (C) $\nabla^2 \vec{A}$
(B) 1 (D) $\vec{A}$

Q.18 The cylindrical coordinates (ρ, θ, z) of the Cartesian point $(1, 1, 1)$ are: [1 Mark]

- (A) $(\sqrt{2}, \pi/4, 1)$ (C) $(\sqrt{2}, \pi/2, 1)$
(B) $(2, \pi/4, 1)$ (D) $(1, \pi/4, 1)$

Q.19 The double integral $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx dy$ evaluates to: [1 Mark]

- (A) $\frac{\pi}{4}$ (C) $\frac{\pi}{2}$
(B) π (D) $\sqrt{\pi}$

Q.20 The angle between the surfaces $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 3$ at $(2, -1, 2)$ is found using their: [1 Mark]

- (A) Gradients (C) Curls
(B) Divergences (D) Laplacians

Q.21 If $f(x, y) = x^3 y + y^3 x$, then $\frac{\partial^2 f}{\partial x \partial y}$ is: [1 Mark]

- (A) $3x^2 + 3y^2$ (C) $x^3 + y^3$
(B) $6xy$ (D) $3x^2 y^2$

Q.22 The value of $\iint_S dS$ over the surface of a sphere of radius a is: [1 Mark]

- (A) $4\pi a^2$ (C) $2\pi a^2$
(B) $\frac{4}{3}\pi a^3$ (D) πa^2

Q.23 If $\vec{F} = x\hat{i} + y\hat{j}$, then $\nabla \cdot \vec{F}$ is: [1 Mark]

- (A) 2 (C) 1
(B) 0 (D) 3

Q.24 The curve C in Stokes' Theorem must be: [1 Mark]

- (A) A closed boundary curve of surface S (C) Infinite
(B) An open straight line (D) Discontinuous

Q.25 The value of $\lim_{(x,y) \rightarrow (0,0)} \frac{x^3+y^3}{x^2+y^2}$ is: [1 Mark]

- (A) 0 (C) Does not exist
(B) 1 (D) ∞

Q.26 The work done by a force $\vec{F}$ along a closed path in a conservative field is: [1 Mark]

- (A) 0 (C) Path dependent
(B) 1 (D) ∞

Q.27 The Jacobian of the inverse transformation satisfies: [1 Mark]

- (A) $J \cdot J' = 1$ (C) $J = J'$
(B) $J + J' = 0$ (D) $J \cdot J' = 0$

Q.28 If $u = f(x/y)$, then $xu_x + yu_y$ is: [1 Mark]

- (A) 0 (C) $x + y$
(B) u (D) 1

Q.29 The volume enclosed by the cylinder $x^2 + y^2 = 1$ between $z = 0$ and $z = 5$ is: [1 Mark]

- (A) 5π (C) 10π
(B) π (D) 25π

Q.30 The Laplacian operator $\nabla^2\phi$ is defined as: [1 Mark]

- (A) $\nabla \cdot (\nabla\phi)$ (C) $\nabla\phi \cdot \vec{r}$
(B) $\nabla \times (\nabla\phi)$ (D) $\|\nabla\phi\|$

PART B: Descriptive Section (Attempt Any 4 of 6) [40 Marks]

Q.31 Verify the **Cayley-Hamilton Theorem** for the
(Unit matrix:
1)

CO1

L3 Apply

10
Marks

$$A = \begin{bmatrix} 1 & 0 & 3 \\ 2 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$$

and hence compute A^{-1} and A^4 .

- Q.32** Solve the second-order non-homogeneous differential equation: **CO2** L3 Apply **10 Marks**
(Unit 2)
- $$(D^2 - 4D + 3)y = 20 \cos 2x + e^x$$

- Q.33** **(a)** Find the radius of convergence and interval of convergence of the power series: **CO3** L4 Analyze **10 Marks**
(Unit 3)

$$\sum_{n=1}^{\infty} \frac{(-1)^n (x-2)^n}{n \cdot 3^n}$$

- (b)** Test the convergence of the series $\sum_{n=1}^{\infty} \left(1 + \frac{1}{\sqrt{n}}\right)^{-n^{3/2}}$ using **Cauchy's Root Test**.

- Q.34** Find all the extreme values (maxima, minima, and saddle points) of the multivariable function: **CO4** L3 Apply **10 Marks**
(Unit 4)
- $$f(x, y) = x^4 + y^4 - 2(x - y)^2$$

- Q.35** Evaluate the double integral by transforming to polar coordinates: **CO5** L3 Apply **10 Marks**
(Unit 5)

$$\iint_R \frac{dx dy}{(1 + x^2 + y^2)^2}$$

where R is the region in the **first quadrant** bounded by the circle $x^2 + y^2 = 1$.

- Q.36** Verify **Stokes' Theorem** for the vector field: **CO6** L4 Analyze **10 Marks**
(Unit 6)

$$\vec{F} = (2x - y)\hat{i} - yz^2\hat{j} - y^2z\hat{k}$$

where S is the upper half surface of the sphere $x^2 + y^2 + z^2 = 1$ ($z \geq 0$) and its boundary curve C is the circle $x^2 + y^2 = 1$ in the xy -plane.

Solutions & Marking Scheme (End-Term Set B)

Part A: Answer Key & Explanations

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	6	A	11	A	16	A	21	A	26	A
2	A	7	A	12	A	17	A	22	A	27	A
3	A	8	A	13	C	18	A	23	A	28	A
4	B	9	A	14	A	19	A	24	A	29	A
5	A	10	A	15	A	20	A	25	A	30	A

Step-by-Step Mathematical Solution Explanations for Critical MCQs

Q.1 (Ans: C) Path $y = mx \implies \frac{1-m^2}{1+m^2}$ depends on slope m . Limit does not exist.

Q.3 (Ans: A) $J = \begin{vmatrix} 2x & -2y \\ 2y & 2x \end{vmatrix} = 4x^2 - (-4y^2) = 4(x^2 + y^2).$

Q.5 (Ans: A) Solenoidal $\implies \nabla \cdot \vec{F} = 1 + 1 + a = 0 \implies a = -2.$

Q.15 (Ans: A) By Green's Theorem, $\oint_C (x dy - y dx) = \iint_R (1 - (-1)) dA = 2 \iint_R dA = 2(\pi \cdot 1^2) = 2\pi.$

Q.19 (Ans: A) $\int_0^{\pi/2} d\theta \int_0^\infty r e^{-r^2} dr = \frac{\pi}{2} \left[-\frac{1}{2} e^{-r^2} \right]_0^\infty = \frac{\pi}{4}.$

Part B: Detailed Subjective Solutions

Step-by-Step Mathematical Solution Solution to Question 31 (10 Marks)

Step 1: Characteristic Equation of A $A = \begin{bmatrix} 1 & 0 & 3 \\ 2 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$.

- $S_1 = 1 + 1 + 1 = 3$.
- $S_2 = M_{11} + M_{22} + M_{33} = (1 - 1) + (1 - 3) + (1 - 0) = 0 - 2 + 1 = -1$.
- $S_3 = \det(A) = 1(0) - 0 + 3(-2 - 1) = -9$.

Equation: $\lambda^3 - 3\lambda^2 - \lambda + 9 = 0$.

Step 2: Verification of Cayley-Hamilton ($A^3 - 3A^2 - A + 9I = O$) Compute A^2 :

$$A^2 = \begin{bmatrix} 1 & 0 & 3 \\ 2 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 3 \\ 2 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 4 & -3 & 6 \\ 3 & 2 & 4 \\ 0 & -2 & 5 \end{bmatrix}$$

Compute $A^3 = A \cdot A^2$:

$$A^3 = \begin{bmatrix} 1 & 0 & 3 \\ 2 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 4 & -3 & 6 \\ 3 & 2 & 4 \\ 0 & -2 & 5 \end{bmatrix} = \begin{bmatrix} 4 & -9 & 21 \\ 11 & -2 & 11 \\ 1 & -7 & 7 \end{bmatrix}$$

Substitute into the polynomial:

$$\begin{aligned} A^3 - 3A^2 - A + 9I &= \begin{bmatrix} 4 & -9 & 21 \\ 11 & -2 & 11 \\ 1 & -7 & 7 \end{bmatrix} - 3 \begin{bmatrix} 4 & -3 & 6 \\ 3 & 2 & 4 \\ 0 & -2 & 5 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 3 \\ 2 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix} + 9 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 4 - 12 - 1 + 9 & -9 + 9 - 0 + 0 & 21 - 18 - 3 + 0 \\ 11 - 9 - 2 + 0 & -2 - 6 - 1 + 9 & 11 - 12 + 1 + 0 \\ 1 - 0 - 1 + 0 & -7 + 6 + 1 + 0 & 7 - 15 - 1 + 9 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = O \end{aligned}$$

Hence, Cayley-Hamilton Theorem is **verified!**

Step 3: Calculating A^{-1} Multiply by A^{-1} : $A^2 - 3A - I + 9A^{-1} = O \implies 9A^{-1} = -A^2 + 3A + I$:

$$9A^{-1} = - \begin{bmatrix} 4 & -3 & 6 \\ 3 & 2 & 4 \\ 0 & -2 & 5 \end{bmatrix} + \begin{bmatrix} 3 & 0 & 9 \\ 6 & 3 & -3 \\ 3 & -3 & 3 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 3 & 3 \\ 3 & 2 & -7 \\ 3 & -1 & -1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{9} \begin{bmatrix} 0 & 3 & 3 \\ 3 & 2 & -7 \\ 3 & -1 & -1 \end{bmatrix}$$

Mark Distribution: • Characteristic equation $\lambda^3 - 3\lambda^2 - \lambda + 9 = 0$: **3 Marks**

• Computation of A^2 , A^3 and verification $= O$: **4 Marks**

• Exact calculation of A^{-1} : **3 Marks**

Step-by-Step Mathematical Solution Solution to Question 32 (10 Marks)

Step 1: Complementary Function $m^2 - 4m + 3 = 0 \implies (m - 1)(m - 3) = 0 \implies m = 1, 3$.

$$y_c = c_1 e^x + c_2 e^{3x}$$

Step 2: Particular Integral for e^x (y_{p_1}) Since $m = 1$ is a single root of the auxiliary equation:

$$y_{p_1} = \frac{1}{(D-1)(D-3)} e^x = \frac{1}{(1-3)} \frac{1}{D-1} e^x = -\frac{1}{2} (x e^x) = -\frac{1}{2} x e^x$$

Step 3: Particular Integral for $20 \cos 2x$ (y_{p_2})

$$y_{p_2} = \frac{1}{D^2 - 4D + 3} (20 \cos 2x)$$

Replace D^2 with $-2^2 = -4$:

$$= \frac{20}{-4 - 4D + 3} \cos 2x = \frac{20}{-4D - 1} \cos 2x = -\frac{20}{4D + 1} \cos 2x$$

Multiply numerator and denominator by $(4D - 1)$:

$$= -20 \frac{4D - 1}{16D^2 - 1} \cos 2x = -20 \frac{4D - 1}{16(-4) - 1} \cos 2x = -20 \frac{4D - 1}{-65} \cos 2x = \frac{4}{13} (4D - 1) \cos 2x$$

Evaluate $D(\cos 2x) = -2 \sin 2x$:

$$= \frac{4}{13} [4(-2 \sin 2x) - \cos 2x] = -\frac{4}{13} (8 \sin 2x + \cos 2x)$$

Step 4: Complete General Solution

$$y(x) = c_1 e^x + c_2 e^{3x} - \frac{1}{2} x e^x - \frac{32}{13} \sin 2x - \frac{4}{13} \cos 2x$$

Mark Distribution: • Complementary function $y_c = c_1 e^x + c_2 e^{3x}$: **2.5 Marks**

• Particular integral y_{p_1} with case of failure: **3.5 Marks**

• Particular integral y_{p_2} for trigonometric term: **3 Marks**

• Final general solution: **1 Mark**

Step-by-Step Mathematical Solution Solution to Question 33 (10 Marks)

Part (a): Radius & Interval of Convergence (5 Marks) Given power series: $\sum_{n=1}^{\infty} c_n(x-2)^n$ where $c_n = \frac{(-1)^n}{n3^n}$.

$$\frac{1}{R} = \lim_{n \rightarrow \infty} \left| \frac{c_{n+1}}{c_n} \right| = \lim_{n \rightarrow \infty} \frac{n \cdot 3^n}{(n+1) \cdot 3^{n+1}} = \frac{1}{3} \implies R = 3$$

The series converges for $|x-2| < 3 \implies -3 < x-2 < 3 \implies -1 < x < 5$. **Testing Endpoints:**

- At $x = 5$: $\sum_{n=1}^{\infty} \frac{(-1)^n(3)^n}{n3^n} = \sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ (Alternating harmonic series). Converges by Leibniz test.
- At $x = -1$: $\sum_{n=1}^{\infty} \frac{(-1)^n(-3)^n}{n3^n} = \sum_{n=1}^{\infty} \frac{(-1)^n(-1)^n3^n}{n3^n} = \sum_{n=1}^{\infty} \frac{1}{n}$ (Harmonic series). Diverges!

Hence, the interval of convergence is $(-1, 5]$.

Part (b): Cauchy Root Test (5 Marks) $u_n = \left(1 + \frac{1}{\sqrt{n}}\right)^{-n^{3/2}}$. Apply Cauchy's Root Test:

$$L = \lim_{n \rightarrow \infty} (u_n)^{1/n} = \lim_{n \rightarrow \infty} \left[\left(1 + \frac{1}{\sqrt{n}}\right)^{-n^{3/2}} \right]^{1/n} = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{\sqrt{n}}\right)^{-n^{1/2}} = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{\sqrt{n}}\right)^{-\sqrt{n}}$$

Let $k = \sqrt{n}$. As $n \rightarrow \infty$, $k \rightarrow \infty$:

$$L = \lim_{k \rightarrow \infty} \left(1 + \frac{1}{k}\right)^{-k} = \frac{1}{e}$$

Since $L = \frac{1}{e} \approx 0.368 < 1$, by Cauchy's Root Test, the series is **CONVERGENT**.

Mark Distribution: • Part (a) Radius $R = 3$: **2 Marks**

- Part (a) Endpoint analysis giving interval $(-1, 5]$: **3 Marks**
- Part (b) Root test power manipulation: **2.5 Marks**
- Part (b) Limit evaluation $1/e < 1$ and conclusion: **2.5 Marks**

Step-by-Step Mathematical Solution Solution to Question 34 (10 Marks)

Step 1: First Partial Derivatives $f(x, y) = x^4 + y^4 - 2(x - y)^2 = x^4 + y^4 - 2x^2 + 4xy - 2y^2$.

$$f_x = 4x^3 - 4x + 4y = 4(x^3 - x + y) = 0 \implies x^3 - x + y = 0$$

$$f_y = 4y^3 + 4x - 4y = 4(y^3 + x - y) = 0 \implies y^3 + x - y = 0$$

Step 2: Finding Critical Points Add (4.2) and (4.2):

$$x^3 + y^3 = 0 \implies y = -x$$

Substitute $y = -x$ into (4.2):

$$x^3 - x - x = 0 \implies x^3 - 2x = 0 \implies x(x^2 - 2) = 0$$

This yields three real solutions:

- $x = 0 \implies y = 0 \implies \mathbf{P_1(0, 0)}$.
- $x = \sqrt{2} \implies y = -\sqrt{2} \implies \mathbf{P_2(\sqrt{2}, -\sqrt{2})}$.
- $x = -\sqrt{2} \implies y = \sqrt{2} \implies \mathbf{P_3(-\sqrt{2}, \sqrt{2})}$.

Step 3: Second Derivative Test

$$r = f_{xx} = 12x^2 - 4, \quad s = f_{xy} = 4, \quad t = f_{yy} = 12y^2 - 4$$

Discriminant: $D = rt - s^2 = (12x^2 - 4)(12y^2 - 4) - 16$.

- At $P_1(0, 0)$: $r = -4, t = -4, s = 4 \implies D = (-4)(-4) - 16 = 0$. Since $D = 0$, the test is inconclusive. Examine behavior along paths: Along $y = x$: $f(x, x) = 2x^4 \geq 0$ (positive near origin). Along $y = 0$: $f(x, 0) = x^4 - 2x^2 = x^2(x^2 - 2) < 0$ for small x . Since f takes both positive and negative values in every neighborhood of $(0, 0)$, $\mathbf{(0, 0)}$ is a **Saddle Point**.
- At $P_2(\sqrt{2}, -\sqrt{2})$: $r = 12(2) - 4 = 20 > 0, t = 12(2) - 4 = 20, D = (20)(20) - 16 = 384 > 0$. Since $D > 0$ and $r > 0$, this is a point of **Local Minimum**. Value: $f(\sqrt{2}, -\sqrt{2}) = 4 + 4 - 2(2\sqrt{2})^2 = 8 - 16 = -8$.
- At $P_3(-\sqrt{2}, \sqrt{2})$: $r = 20 > 0, t = 20, D = 384 > 0$. This is also a point of **Local Minimum**. Value: $f(-\sqrt{2}, \sqrt{2}) = -8$.

Mark Distribution: • First partial derivatives and critical points $(0, 0), (\pm\sqrt{2}, \mp\sqrt{2})$: **3.5 Marks**

• Second derivative test discriminant: **2.5 Marks**

• Investigating $(0, 0)$ saddle point: **2 Marks**

• Classifying local minima at $(\pm\sqrt{2}, \mp\sqrt{2})$ with value -8 : **2 Marks**

Step-by-Step Mathematical Solution Solution to Question 35 (10 Marks)

Step 1: Polar Coordinate Transformation Let $x = r \cos \theta$, $y = r \sin \theta \implies x^2 + y^2 = r^2$, and $dx dy = r dr d\theta$. The region R is the first quadrant of the unit disk $x^2 + y^2 \leq 1$:

$$r \in [0, 1], \quad \theta \in \left[0, \frac{\pi}{2}\right]$$

Step 2: Transformed Integral

$$I = \int_0^{\pi/2} \int_0^1 \frac{r dr d\theta}{(1+r^2)^2} = \left(\int_0^{\pi/2} d\theta \right) \left(\int_0^1 \frac{r dr}{(1+r^2)^2} \right)$$

Step 3: Evaluation Angular integral: $\int_0^{\pi/2} d\theta = \frac{\pi}{2}$. Radial integral: Let $u = 1 + r^2 \implies du = 2r dr \implies r dr = \frac{du}{2}$. When $r = 0, u = 1$; when $r = 1, u = 2$:

$$\int_0^1 \frac{r dr}{(1+r^2)^2} = \frac{1}{2} \int_1^2 u^{-2} du = \frac{1}{2} \left[-\frac{1}{u} \right]_1^2 = \frac{1}{2} \left(-\frac{1}{2} - (-1) \right) = \frac{1}{2} \left(\frac{1}{2} \right) = \frac{1}{4}$$

Multiply both factors:

$$I = \left(\frac{\pi}{2} \right) \left(\frac{1}{4} \right) = \frac{\pi}{8}$$

Mark Distribution: • Polar coordinate conversion with Jacobian r : **3 Marks**

- Limits of integration for first quadrant: **2 Marks**
- Substitution $u = 1 + r^2$ and integration: **3 Marks**
- Final exact answer $\pi/8$: **2 Marks**

Step-by-Step Mathematical Solution Solution to Question 36 (10 Marks)**Step 1: Statement of Stokes' Theorem**

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \hat{n} dS$$

Given $\vec{F} = (2x - y)\hat{i} - yz^2\hat{j} - y^2z\hat{k}$ over the hemisphere $x^2 + y^2 + z^2 = 1$ ($z \geq 0$). The boundary curve C is the circle $x^2 + y^2 = 1, z = 0$.

Step 2: Line Integral Evaluation ($\oint_C \vec{F} \cdot d\vec{r}$) On the boundary circle C in the xy -plane: $z = 0, dz = 0$. Parameterized by $x = \cos \theta, y = \sin \theta$ for $\theta \in [0, 2\pi]$:

$$dx = -\sin \theta d\theta, \quad dy = \cos \theta d\theta$$

Since $z = 0$, the $\hat{j}$ and $\hat{k}$ components of $\vec{F}$ vanish identically!

$$\vec{F} \cdot d\vec{r} = (2x - y)dx = (2\cos \theta - \sin \theta)(-\sin \theta d\theta) = (-2\sin \theta \cos \theta + \sin^2 \theta)d\theta$$

$$\oint_C \vec{F} \cdot d\vec{r} = \int_0^{2\pi} (-\sin 2\theta + \sin^2 \theta)d\theta = 0 + \int_0^{2\pi} \left(\frac{1 - \cos 2\theta}{2} \right) d\theta = \pi$$

Step 3: Surface Integral Evaluation ($\iint_S (\nabla \times \vec{F}) \cdot \hat{n} dS$) Compute $\nabla \times \vec{F}$:

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x - y & -yz^2 & -y^2z \end{vmatrix}$$

- $\hat{i}$ -component: $\frac{\partial}{\partial y}(-y^2z) - \frac{\partial}{\partial z}(-yz^2) = -2yz - (-2yz) = 0$.
- $\hat{j}$ -component: $\frac{\partial}{\partial z}(2x - y) - \frac{\partial}{\partial x}(-y^2z) = 0 - 0 = 0$.
- $\hat{k}$ -component: $\frac{\partial}{\partial x}(-yz^2) - \frac{\partial}{\partial y}(2x - y) = 0 - (-1) = 1$.

Thus:

$$\nabla \times \vec{F} = 0\hat{i} + 0\hat{j} + 1\hat{k} = \hat{k}$$

Now by Stokes' Theorem, we can choose the planar base surface S_1 (the circular disk $x^2 + y^2 \leq 1, z = 0$) because both share the identical boundary curve C ! On S_1 , $\hat{n} = \hat{k}$:

$$(\nabla \times \vec{F}) \cdot \hat{n} = \hat{k} \cdot \hat{k} = 1$$

$$\iint_S (\nabla \times \vec{F}) \cdot \hat{n} dS = \iint_{S_1} 1 dS = \text{Area of unit circle} = \pi(1^2) = \pi$$

Since Line Integral (π) = Surface Integral (π), **Stokes' Theorem is verified!**

Mark Distribution: • Line integral evaluation yielding π : **4 Marks**

• Computing $\nabla \times \vec{F} = \hat{k}$: **3 Marks**

• Surface integral evaluation yielding π : **2 Marks**

• Equating both and concluding verification: **1 Mark**

4.3 End-Term Examination — Set C

LOVELY PROFESSIONAL UNIVERSITY

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Course Code: MTH-165 **Course Title:** Mathematics for Engineers

Academic Task Type: End-Term Examination **Set:** C

Maximum Marks: 60 Marks **Time Allowed:** 3 Hours

Course Outcomes Addressed: CO1 to CO6 (Units 1 to 6 Comprehensive)

Roll No.:

Reg. No.:

Section:

Structure of Question Paper:

Part A (Objective): 30 MCQs strictly from **Units 4, 5, and 6**. Total weight: **20 Marks**. All questions compulsory.

Part B (Subjective): 6 Questions from **all units** (10 Marks each). **Attempt any 4 questions** ($4 \times 10 = 40$ Marks).

Total Maximum Marks: 60 Marks.

PART A: Objective Section (30 MCQs — Units 4, 5, 6) [20 Marks]

Q.1 The value of $\lim_{(x,y) \rightarrow (0,0)} \frac{x^4 y^4}{(x^2 + y^4)^3}$ along the path $x = y^2$ is:

[1 Mark]

- (A) $1/8$ (C) 0
(B) $1/4$ (D) 1

Q.2 If $u = \tan^{-1} \left(\frac{x^3 + y^3}{x - y} \right)$, then $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}$ is equal to:

[1 Mark]

- (A) $\sin 2u$ (C) $2 \tan u$
(B) $\cos 2u$ (D) $\tan 2u$

Q.3 The saddle point of $f(x, y) = 2x^2 - xy - 3y^2 - 3x + 7y$ is:

[1 Mark]

- (A) (1, 1) (C) (1, -1)
(B) (0, 0) (D) (2, 1)

Q.4 The directional derivative of $\phi = 2xy + z^2$ at $(1, -1, 3)$ in the direction of $2\hat{i} + 2\hat{j} - \hat{k}$ is:

[1 Mark]

- (A) $-14/3$ (C) $14/3$
(B) $-10/3$ (D) 4

Q.5 If $\vec{F} = (x + 2y)\hat{i} + (y + 3z)\hat{j} + (z + cx)\hat{k}$ is solenoidal, then c can be:

[1 Mark]

- (A) Any value because $\nabla \cdot \vec{F}$ is independent of c (C) -1
(B) 0 (D) -3

Q.6 The curl of the vector field $\vec{F} = x^2\hat{i} + y^2\hat{j} + z^2\hat{k}$ is: [1 Mark]

- (A) $\vec{0}$ (C) $2(x + y + z)$
(B) $2x\hat{i} + 2y\hat{j} + 2z\hat{k}$ (D) $\vec{r}$

Q.7 The value of $\int_0^1 \int_0^x (x^2 + y^2) dy dx$ is: [1 Mark]

- (A) $1/3$ (C) $1/4$
(B) $1/6$ (D) $5/12$

Q.8 In changing from Cartesian to spherical coordinates, the Jacobian determinant $|J|$ is: [1 Mark]

- (A) $\rho^2 \sin \phi$ (C) $\rho^2 \cos \phi$
(B) $\rho \sin \phi$ (D) $\rho \cos \phi$

Q.9 If $\nabla \times \vec{F} = \vec{0}$, the line integral $\int_C \vec{F} \cdot d\vec{r}$ is: [1 Mark]

- (A) Path independent (C) 0 always
(B) Path dependent (D) ∞

Q.10 The value of $\nabla^2(r^3)$ where $r = \|\vec{r}\|$ is: [1 Mark]

- (A) $12r$ (C) $3r$
(B) $6r$ (D) $12r^2$

Q.11 The value of $\oint_C \vec{r} \cdot d\vec{r}$ around any closed smooth curve C is: [1 Mark]

- (A) 0 (C) π
(B) 2π (D) Enclosed area

Q.12 The function $f(x, y) = x^2 - y^2$ has at $(0, 0)$: [1 Mark]

- (A) A local minimum (C) A saddle point
(B) A local maximum (D) No critical point

Q.13 The value of $\int_0^2 \int_0^1 (xy + e^x) dy dx$ is: [1 Mark]

- (A) $e^2 + 1$ (C) e^2
(B) $e^2 - 1$ (D) $2e^2$

Q.14 A surface integral $\iint_S \vec{F} \cdot \hat{n} dS$ represents the: [1 Mark]

- (A) Total flux of $\vec{F}$ through S (C) Circulation
(B) Work done (D) Volume

Q.15 If $\vec{F} = y\hat{i} - x\hat{j}$, then $\nabla \times \vec{F}$ is: [1 Mark]

- (A) $-2\hat{k}$ (C) $\vec{0}$
(B) $2\hat{k}$ (D) $\hat{k}$

Q.16 The area enclosed by the astroid $x^{2/3} + y^{2/3} = a^{2/3}$ is given by: [1 Mark]

- (A) $\frac{3}{8}\pi a^2$ (C) πa^2
(B) $\frac{3}{4}\pi a^2$ (D) $\frac{1}{2}\pi a^2$

Q.17 The value of $\iint_S \hat{n} dS$ over any closed surface S is identically: [1 Mark]

- (A) $\vec{0}$ (C) Surface Area
(B) 1 (D) $V\hat{k}$

Q.18 The stationary value of $x^2 + y^2 + z^2$ subject to $x + y + z = 3$ is: [1 Mark]

- (A) 3 (C) 1
(B) 9 (D) 6

Q.19 If u, v are functionally dependent, their Jacobian $\frac{\partial(u,v)}{\partial(x,y)}$ is: [1 Mark]

- (A) 0 (C) -1
(B) 1 (D) ∞

Q.20 The volume of the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \leq 1$ is: [1 Mark]

- (A) $\frac{4}{3}\pi abc$ (C) $\frac{2}{3}\pi abc$
(B) $4\pi abc$ (D) πabc

Q.21 The value of $\nabla(\vec{a} \cdot \vec{r})$ where $\vec{a}$ is a constant vector is: [1 Mark]

- (A) $\vec{a}$ (C) $2\vec{a}$
(B) $\vec{0}$ (D) $\vec{r}$

Q.22 The value of the integral $\int_0^1 \int_0^{\sqrt{1-x^2}} dy dx$ is: [1 Mark]

- (A) $\frac{\pi}{4}$ (C) π
(B) $\frac{\pi}{2}$ (D) 1

Q.23 If $f(x, y)$ has continuous partial derivatives, then $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$ is guaranteed by: [1 Mark]

- (A) Schwarz's Theorem (C) Taylor's Theorem
(B) Euler's Theorem (D) Green's Theorem

Q.24 The divergence of $\vec{r}/r^3$ for $r \neq 0$ is: [1 Mark]

- (A) 0 (C) $-3/r^4$
(B) $3/r^3$ (D) $1/r^2$

Q.25 The double integral $\iint_R dx dy$ over $0 \leq x \leq 2, 0 \leq y \leq 3$ is: [1 Mark]

- (A) 6 (C) 1
(B) 5 (D) 12

Q.26 The unit outward normal to the sphere $x^2 + y^2 + z^2 = R^2$ at any point is: [1 Mark]

- (A) $\frac{\vec{r}}{R}$ (C) $\frac{\nabla r}{R}$
(B) $\frac{\vec{r}}{R^2}$ (D) $\vec{r}$

Q.27 The value of $\oint_C \nabla \phi \cdot d\vec{r}$ around any closed curve is: [1 Mark]

- (A) 0 (C) 2π
(B) ϕ (D) 1

Q.28 If $u = x + y + z, v = xy + yz + zx, w = x^2 + y^2 + z^2$, the Jacobian $\frac{\partial(u,v,w)}{\partial(x,y,z)}$ is: [1 Mark]

- (A) 0 (C) 2
(B) 1 (D) 6

Q.29 The volume of a right circular cone of base radius R and height h is: [1 Mark]

- (A) $\frac{1}{3}\pi R^2 h$ (C) $\frac{2}{3}\pi R^2 h$
(B) $\pi R^2 h$ (D) $\frac{1}{2}\pi R^2 h$

Q.30 If $\vec{F} = \text{curl}(\vec{A})$, then the flux of $\vec{F}$ across any closed surface S is: [1 Mark]

- (A) 0 (C) Area of S
(B) 1 (D) Volume of S
-

PART B: Descriptive Section (Attempt Any 4 of 6) [40 Marks]

- Q.31** Diagonalize the real symmetric matrix A by an orthogonal transformation P : **CO1** L3 Apply **10 Marks**
(Unit 1)

$$A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$$

Find the modal matrix P and the resulting diagonal matrix $D = P^T A P$.

- Q.32** Using the **Method of Variation of Parameters**, solve the second-order ordinary differential equation: **CO2** L4 Analyze **10 Marks**
(Unit 2)

$$\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} + y = \frac{e^x}{x} \quad (x > 0)$$

- Q.33** (a) Test the convergence of the infinite series: **CO3** L3 Apply **10 Marks**
(Unit 3)
- $$\sum_{n=1}^{\infty} \frac{n^3 + a}{2^n + b} \quad (a, b > 0)$$

(b) Expand $f(x) = e^x \sin x$ in a **Maclaurin series** up to the term containing x^5 .

- Q.34** Find the maximum and minimum values of the function: **CO4** L3 Apply **10 Marks**
(Unit 4)
- $$f(x, y, z) = xyz$$

subject to the spherical constraint $x^2 + y^2 + z^2 = 1$, using **Lagrange Multipliers**.

- Q.35** Evaluate the triple integral: **CO5** L3 Apply **10 Marks**
(Unit 5)
- $$\iiint_V xyz \, dx \, dy \, dz$$

over the **positive octant** of the ellipsoid:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \leq 1$$

Q.36 Verify **Green's Theorem in the plane** for:
(Unit 6)

CO6

L4 Analyze

10
Marks

$$\oint_C [(xy + y^2) dx + x^2 dy]$$

where C is the closed curve bounding the region between the line $y = x$ and the parabola $y = x^2$.

Solutions & Marking Scheme (End-Term Set C)

Part A: Answer Key & Explanations

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	6	A	11	A	16	A	21	A	26	A
2	A	7	A	12	C	17	A	22	A	27	A
3	A	8	A	13	A	18	A	23	A	28	A
4	A	9	A	14	A	19	A	24	A	29	A
5	A	10	A	15	A	20	A	25	A	30	A

Step-by-Step Mathematical Solution Explanations for Part A

Q.1 (Ans: A) Along $x = y^2$: $\frac{(y^2)^4 y^4}{((y^2)^2 + y^4)^3} = \frac{y^{12}}{(2y^4)^3} = \frac{y^{12}}{8y^{12}} = \frac{1}{8}$.

Q.2 (Ans: A) Let $f = \tan u = \frac{x^3 + y^3}{x - y}$ which is homogeneous of degree 2. By Euler's Theorem, $xf_x + yf_y = 2f \implies \sec^2 u(xu_x + yu_y) = 2 \tan u \implies xu_x + yu_y = \frac{2 \tan u}{\sec^2 u} = 2 \sin u \cos u = \sin 2u$.

Q.10 (Ans: A) $\nabla^2(r^n) = n(n+1)r^{n-2}$. For $n = 3$: $3(4)r^{3-2} = 12r$.

Q.24 (Ans: A) $\nabla \cdot (\vec{r}/r^3) = \frac{\nabla \cdot \vec{r}}{r^3} + \vec{r} \cdot \nabla(r^{-3}) = \frac{3}{r^3} + \vec{r} \cdot (-3r^{-5}\vec{r}) = \frac{3}{r^3} - \frac{3r^2}{r^5} = 0$.

Q.28 (Ans: A) Since $w = x^2 + y^2 + z^2 = (x + y + z)^2 - 2(xy + yz + zx) = u^2 - 2v$, w is a functional combination of u and v . Hence the Jacobian vanishes identically ($J = 0$).

Q.30 (Ans: A) By Gauss Divergence Theorem, $\iint_S (\nabla \times \vec{A}) \cdot \hat{n} dS = \iiint_V \nabla \cdot (\nabla \times \vec{A}) dV = \iiint_V 0 dV = 0$.

Part B: Detailed Subjective Solutions

Step-by-Step Mathematical Solution Solution to Question 31 (10 Marks)

Step 1: Eigenvalues of A $A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$.

- $S_1 = 3 + 5 + 3 = 11$.
- $S_2 = (15 - 1) + (9 - 1) + (15 - 1) = 14 + 8 + 14 = 36$.
- $S_3 = \det(A) = 3(14) - (-1)(-3 + 1) + 1(1 - 5) = 42 - 2 - 4 = 36$.

Equation: $\lambda^3 - 11\lambda^2 + 36\lambda - 36 = 0$. Factorization: $(\lambda - 2)(\lambda - 3)(\lambda - 6) = 0 \implies \lambda_1 = 2, \lambda_2 = 3, \lambda_3 = 6$.

Step 2: Normalized Eigenvectors

- For $\lambda_1 = 2$: $(A - 2I)X = 0 \implies \begin{bmatrix} 1 & -1 & 1 \\ -1 & 3 & -1 \\ 1 & -1 & 1 \end{bmatrix} X = 0$. $x_1 - x_2 + x_3 = 0$ and $-x_1 +$

$$3x_2 - x_3 = 0 \implies 2x_2 = 0 \implies x_2 = 0, x_3 = -x_1. \text{ Eigenvector } X_1 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}. \text{ Normalized:}$$

$$\hat{X}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}.$$

- For $\lambda_2 = 3$: $(A - 3I)X = 0 \implies \begin{bmatrix} 0 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 0 \end{bmatrix} X = 0 \implies x_1 = x_2 = x_3$. Eigenvector

$$X_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}. \text{ Normalized: } \hat{X}_2 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

- For $\lambda_3 = 6$: $(A - 6I)X = 0 \implies \begin{bmatrix} -3 & -1 & 1 \\ -1 & -1 & -1 \\ 1 & -1 & -3 \end{bmatrix} X = 0$. Cross-multiplication gives

$$x_1 = 1, x_2 = -2, x_3 = 1. \text{ Eigenvector } X_3 = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}. \text{ Normalized: } \hat{X}_3 = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}.$$

Step 3: Orthogonal Modal Matrix P and Diagonal Matrix D

$$P = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{3} & 1/\sqrt{6} \\ 0 & 1/\sqrt{3} & -2/\sqrt{6} \\ -1/\sqrt{2} & 1/\sqrt{3} & 1/\sqrt{6} \end{bmatrix}$$

Since P is orthogonal ($P^{-1} = P^T$):

$$D = P^T A P = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 6 \end{bmatrix}$$

Mark Distribution: • Characteristic equation and eigenvalues $\lambda = 2, 3, 6$: **3 Marks**

• Eigenvectors determination: **3 Marks**

• Normalization of eigenvectors: **2 Marks**

• Construction of modal matrix P and diagonal matrix D : **2 Marks**

Step-by-Step Mathematical Solution Solution to Question 32 (10 Marks)

Step 1: Complementary Function $m^2 - 2m + 1 = 0 \implies (m - 1)^2 = 0 \implies m = 1, 1.$

$$y_1 = e^x, \quad y_2 = xe^x \implies \mathbf{y}_c = (\mathbf{c}_1 + \mathbf{c}_2 x)e^x$$

Step 2: Wronskian $W(y_1, y_2)$

$$W = \begin{vmatrix} e^x & xe^x \\ e^x & (x+1)e^x \end{vmatrix} = e^{2x}(x+1) - xe^{2x} = e^{2x} \neq 0$$

Step 3: Particular Integral via Variation of Parameters Let $y_p = u(x)y_1 + v(x)y_2$ where $R(x) = \frac{e^x}{x}$:

$$u(x) = - \int \frac{y_2 R(x)}{W} dx = - \int \frac{(xe^x)(e^x/x)}{e^{2x}} dx = - \int 1 dx = -x$$

$$v(x) = \int \frac{y_1 R(x)}{W} dx = \int \frac{(e^x)(e^x/x)}{e^{2x}} dx = \int \frac{1}{x} dx = \ln x$$

Step 4: Particular Integral and General Solution

$$y_p = (-x)e^x + (\ln x)(xe^x) = xe^x(\ln x - 1)$$

Absorbing the $-xe^x$ term into the arbitrary constant c_2xe^x of y_c :

$$\mathbf{y}(x) = \mathbf{c}_1 e^x + \mathbf{c}_2 x e^x + x e^x \ln x$$

Mark Distribution: • Complementary function $y_c = (c_1 + c_2 x)e^x$: **2 Marks**

• Wronskian computation $W = e^{2x}$: **2 Marks**

• Integrals $u(x) = -x$ and $v(x) = \ln x$: **4 Marks**

• Final general solution: **2 Marks**

Step-by-Step Mathematical Solution Solution to Question 33 (10 Marks)

Part (a): Convergence of Series (5 Marks) Let $u_n = \frac{n^3+a}{2^n+b}$. As $n \rightarrow \infty$, $u_n \approx \frac{n^3}{2^n}$. Apply D'Alembert's Ratio Test:

$$L = \lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{(n+1)^3 + a}{2^{n+1} + b} \cdot \frac{2^n + b}{n^3 + a} = \lim_{n \rightarrow \infty} \frac{(n+1)^3}{n^3} \cdot \frac{2^n}{2^{n+1}} = (1) \cdot \frac{1}{2} = \frac{1}{2}$$

Since $L = \frac{1}{2} < 1$, the series is **strictly CONVERGENT**.

Part (b): Maclaurin Series of $e^x \sin x$ (5 Marks) We know the standard Maclaurin series:

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \dots$$

$$\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \dots$$

Multiply the series term by term up to degree 5:

$$e^x \sin x = \left(1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24}\right) \left(x - \frac{x^3}{6} + \frac{x^5}{120}\right)$$

- Degree 1: $1 \cdot x = x$
- Degree 2: $x \cdot x = x^2$
- Degree 3: $1 \cdot (-x^3/6) + \frac{x^2}{2} \cdot x = -\frac{1}{6}x^3 + \frac{1}{2}x^3 = \frac{1}{3}x^3$
- Degree 4: $x \cdot (-x^3/6) + \frac{x^3}{6} \cdot x = -\frac{1}{6}x^4 + \frac{1}{6}x^4 = 0$
- Degree 5: $1 \cdot (x^5/120) + \frac{x^2}{2}(-x^3/6) + \frac{x^4}{24}(x) = \left(\frac{1}{120} - \frac{1}{12} + \frac{1}{24}\right)x^5 = \frac{1-10+5}{120}x^5 = -\frac{4}{120}x^5 = -\frac{1}{30}x^5$

$$e^x \sin x = x + x^2 + \frac{1}{3}x^3 - \frac{1}{30}x^5 + \dots$$

Mark Distribution: • Part (a) Ratio test limit $L = 1/2 < 1$ (Convergent): **5 Marks**
• Part (b) Series expansion multiplication up to degree 5: **5 Marks**

Step-by-Step Mathematical Solution Solution to Question 34 (10 Marks)

Step 1: Lagrange Equations Objective: $f(x, y, z) = xyz$. Constraint: $g(x, y, z) = x^2 + y^2 + z^2 - 1 = 0$. Set $\nabla f = \lambda \nabla g$:

$$yz = 2\lambda x \quad (4.4)$$

$$xz = 2\lambda y \quad (4.5)$$

$$xy = 2\lambda z \quad (4.6)$$

Step 2: Solving the System Multiply (4.4) by x , (4.5) by y , and (4.6) by z :

$$xyz = 2\lambda x^2 = 2\lambda y^2 = 2\lambda z^2$$

If $xyz \neq 0$, then $\lambda \neq 0$, which immediately implies:

$$x^2 = y^2 = z^2$$

Substitute into the constraint $x^2 + y^2 + z^2 = 1$:

$$3x^2 = 1 \implies x^2 = \frac{1}{3} \implies x = \pm \frac{1}{\sqrt{3}}, \quad y = \pm \frac{1}{\sqrt{3}}, \quad z = \pm \frac{1}{\sqrt{3}}$$

Step 3: Extreme Values

- When all three coordinates have positive signs (or two negative, one positive):

$$f_{\max} = \left(\frac{1}{\sqrt{3}}\right) \left(\frac{1}{\sqrt{3}}\right) \left(\frac{1}{\sqrt{3}}\right) = \frac{1}{3\sqrt{3}}$$

- When all three are negative (or one negative, two positive):

$$f_{\min} = -\frac{1}{3\sqrt{3}}$$

Mark Distribution: • Setting up Lagrange system: **3 Marks**

• Deducing $x^2 = y^2 = z^2 = 1/3$: **4 Marks**

• Finding $f_{\max} = \frac{1}{3\sqrt{3}}$ and $f_{\min} = -\frac{1}{3\sqrt{3}}$: **3 Marks**

Step-by-Step Mathematical Solution Solution to Question 35 (10 Marks)**Step 1: Transformation to Unit Sphere** Substitute:

$$x = au, \quad y = bv, \quad z = cw$$

Then the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \leq 1$ transforms into the unit sphere:

$$u^2 + v^2 + w^2 \leq 1, \quad u \geq 0, v \geq 0, w \geq 0$$

The Jacobian of the transformation is:

$$J = \frac{\partial(x, y, z)}{\partial(u, v, w)} = abc \implies dx dy dz = abc du dv dw$$

Step 2: Integral in (u, v, w) Space

$$I = \iiint_V (au)(bv)(cw)(abc du dv dw) = a^2 b^2 c^2 \iiint_{V'} uvw du dv dw$$

Step 3: Spherical Polar Coordinates in (u, v, w) Let $u = \rho \sin \phi \cos \theta$, $v = \rho \sin \phi \sin \theta$, $w = \rho \cos \phi$. For the positive octant: $\rho \in [0, 1]$, $\phi \in [0, \pi/2]$, $\theta \in [0, \pi/2]$:

$$\iiint_{V'} uvw du dv dw = \int_0^{\pi/2} \cos \theta \sin \theta d\theta \int_0^{\pi/2} \sin^3 \phi \cos \phi d\phi \int_0^1 \rho^5 d\rho$$

- $\int_0^{\pi/2} \sin \theta \cos \theta d\theta = \left[\frac{\sin^2 \theta}{2} \right]_0^{\pi/2} = \frac{1}{2}$
- $\int_0^{\pi/2} \sin^3 \phi \cos \phi d\phi = \left[\frac{\sin^4 \phi}{4} \right]_0^{\pi/2} = \frac{1}{4}$
- $\int_0^1 \rho^5 d\rho = \frac{1}{6}$

Multiplying:

$$\frac{1}{2} \times \frac{1}{4} \times \frac{1}{6} = \frac{1}{48}$$

Therefore:

$$I = \frac{a^2 b^2 c^2}{48}$$

Mark Distribution: • Ellipsoid transformation and Jacobian abc : **3 Marks**

- Spherical coordinate conversion for positive octant: **3 Marks**
- Evaluation of individual angular and radial integrals: **3 Marks**
- Final answer $a^2 b^2 c^2 / 48$: **1 Mark**

Step-by-Step Mathematical Solution Solution to Question 36 (10 Marks)**Step 1: Statement of Green's Theorem**

$$\oint_C (Pdx + Qdy) = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy$$

Here $P = xy + y^2$ and $Q = x^2$. Boundaries: $y = x^2$ and $y = x$. Intersections: $x^2 = x \implies x = 0$ and $x = 1$. Vertices: $(0, 0)$ and $(1, 1)$.

Step 2: Double Integral Evaluation

$$\frac{\partial Q}{\partial x} = 2x, \quad \frac{\partial P}{\partial y} = x + 2y$$

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 2x - (x + 2y) = x - 2y$$

Set up the double integral:

$$\begin{aligned} \iint_R (x - 2y) dy dx &= \int_0^1 \int_{x^2}^x (x - 2y) dy dx = \int_0^1 [xy - y^2]_{x^2}^x dx \\ &= \int_0^1 [(x^2 - x^2) - (x^3 - x^4)] dx = \int_0^1 (x^4 - x^3) dx = \left[\frac{x^5}{5} - \frac{x^4}{4} \right]_0^1 = \frac{1}{5} - \frac{1}{4} = -\frac{1}{20} \end{aligned}$$

Step 3: Line Integral Evaluation ($\oint_C Pdx + Qdy$) The curve $C = C_1 \cup C_2$:

(i) Along $C_1 : y = x^2$ from $(0, 0)$ to $(1, 1)$, $dy = 2xdx$:

$$\int_{C_1} = \int_0^1 [(x(x^2) + (x^2)^2)dx + x^2(2xdx)] = \int_0^1 (x^3 + x^4 + 2x^3)dx = \int_0^1 (3x^3 + x^4)dx = \frac{3}{4} + \frac{1}{5} = \frac{19}{20}$$

(ii) Along $C_2 : y = x$ from $(1, 1)$ to $(0, 0)$, $dy = dx$:

$$\int_{C_2} = \int_1^0 [(x^2 + x^2)dx + x^2dx] = \int_1^0 3x^2dx = [x^3]_1^0 = 0 - 1 = -1$$

Total line integral:

$$\oint_C = \int_{C_1} + \int_{C_2} = \frac{19}{20} - 1 = -\frac{1}{20}$$

Since Line Integral $(-1/20) =$ Double Integral $(-1/20)$, **Green's Theorem is verified!**

Mark Distribution: • Stating Green's Theorem and computing $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$: **2 Marks**

• Evaluating double integral yielding $-1/20$: **3 Marks**

• Evaluating line integrals along C_1 and C_2 : **4 Marks**

• Equating results and concluding verification: **1 Mark**